CHAPTER-1
INTRODUCTION

1.1 OVERVIEW

The Tool Request Management System is a cloud-based application developed using the salesforce platform. the project uses salesforce technologies such as experience cloud, apex controllers, lightning web components (LWC), flows, reports, and dashboards to manage and track laboratory tool requests in an educational institution. The main aim of the system is to provide a centralized and automated platform that simplifies the process of requesting, approving, and managing laboratory tools for students and lab staff.

In many institutions, laboratory tools are often issued and tracked manually using register books, spreadsheets, or verbal requests, which can cause errors and delays. The tool request management system solves these problems by providing a digital platform that records and monitors tool requests efficiently. students access the experience cloud portal as guest users, complete the student registration form, and receive an application number that is used to submit tool requests.

After registration, students submit a tool request form by selecting the required tool and quantity. Once the request is submitted, it is stored in salesforce and an email notification is sent to the lab incharge for review. If the lab incharge approves the request, an email notification is sent to the head of department for final approval. the system also provides reports and dashboards to monitor tool requests and manage laboratory resources efficiently.

The system updates the request status after each stage of approval. students can view the current status of their requests through the portal. This helps maintain transparency in the tool request process. It also allows laboratory authorities to track and manage tool usage effectively. The system also sends notifications to the concerned authorities during the approval process. This ensures that requests are processed quickly and efficiently. The system also maintains proper records of all requests in the database. This helps administrators review and manage tool requests easily in the future.

1.2 EXISTING SYSTEM

In many educational institutions, the process of issuing and managing laboratory tools is usually handled using manual met head of departments. Students typically request tools directly from the lab staff or write their requirements in a register. The lab staff then checks the availability of tools and records the details manually, which can take time and requires continuous monitoring.

The existing system commonly depends on register books, spreadsheets, or paper-based records to store and track tool request details. Lab staff must manually record student information, the tools requested, and the quantity issued. This manual process increases the chances of errors such as incorrect entries, missing information, or difficulty in maintaining proper records, especially when many students request tools at the same time.

Another limitation of the existing system is the lack of automation and structured approval processes. Tool requests are often handled directly by lab staff without proper tracking or approval from higher authorities such as the head of department. Additionally, there is no automated notification system to inform lab staff or department heads about tool requests, making it difficult to monitor tool usage and manage laboratory resources efficiently.

1.3 DRAWBACKS OF EXISTING SYSTEM

- The system mainly depends on manual met head of departments such as registers or spreadsheets to record tool requests.
- There is no centralized digital system to track laboratory tool usage.
- Manual data entry may lead to errors such as duplicate entries, missing information, or incorrect records.
- It becomes difficult to monitor tool availability and usage when many students request tools at the same time.
- The existing system does not have a structured approval process involving lab incharge and head of department.
- The system lacks transparency and proper monitoring of laboratory resources.
- It is difficult for students to track the status of their tool requests.

1.4 PROPOSED SYSTEM

The proposed tool request management system is a cloud-based solution developed using the salesforce platform to automate and simplify the management of laboratory tool requests. in this system, students can access the experience cloud portal and complete the student registration form. After registration, the system generates a unique application number, which students use to submit a tool request form by selecting the required tool and quantity. the request details are stored in salesforce, allowing the system to maintain proper records of all tool requests.

The proposed system also includes an automated approval process. Once a tool request is submitted, an email notification is sent to the lab incharge to review the request. If the Lab Incharge approves it, the request is forwarded to the Head of Department (HOD) for final approval. This automated workflow improves efficiency, reduces manual work, and ensures better monitoring and management of laboratory tools within the institution.

1.5 ADVANTAGES OF PROPOSED SYSTEM

- Provides a centralized system to manage laboratory tool requests.
- Reduces manual work by automating the request and approval process.
- Allows students to easily register and submit tool requests online.
- Sends automatic email notifications to lab incharge and head of department.
- Maintains accurate records of all tool requests.

SUMMARY

The tool request management system is developed using the salesforce platform to automate the process of managing laboratory tool requests. the system allows students to register and submit tool requests through the experience cloud portal, and the requests are stored in salesforce for proper record management. after submission, the request is first reviewed by the lab incharge, and if approved, it is forwarded to the head of department head of department for final approval, with email notifications sent at each stage. This system reduces manual work, improves efficiency, and ensures better monitoring and management of laboratory tools. It also helps maintain accurate records of all tool requests in the system. Students can easily track the status of their requests through the portal.

CHAPTER-2
SYSTEM ANALYSIS

2.1 IDENTIFICATION OF NEED

The need for the tool request management system arises from the difficulties faced in managing laboratory tools using manual met head of departments. in many institutions, tool requests are recorded in registers or spreadsheets, making it difficult to track tool usage and maintain accurate records. This manual process can lead to errors, delays, and inefficient management of laboratory resources.

To overcome these challenges, there is a need for an automated system that can manage tool requests in a structured and organized way. The proposed system helps students easily submit tool requests and enables the lab incharge and head of department to review and approve them efficiently. It also allows the system to store request details securely and maintain proper records. The system provides better transparency and monitoring of laboratory tools. Automated notifications help authorities respond quickly to requests. Overall, the system improves efficiency and ensures effective management of laboratory resources.

The system also helps reduce the manual workload of laboratory staff by automating the request and approval process. It ensures that all tool requests are properly recorded and can be easily accessed whenever needed. This improves the accuracy of information and helps maintain organized records within the system. In addition, the system provides a centralized platform where students and laboratory authorities can interact easily. It improves communication between students, lab incharge, and head of department by providing timely updates about tool requests and approvals. This ensures smooth coordination and effective utilization of laboratory tools. The system also supports better decision-making through reports and dashboards. These reports help administrators monitor tool usage and request trends. This helps institutions manage laboratory resources more efficiently.

2.2 FEASIBILITY STUDY

A feasibility study is conducted to evaluate whether the proposed tool request management system can be successfully developed and implemented. It ensures that the system is practical, easy to use, and beneficial for managing laboratory tool requests. In this project, three types of feasibility are considered:

- Technical Feasibility
- Operational Feasibility
- Economic Feasibility

2.2.1 TECHNICAL FEASIBILITY

Technical feasibility refers to evaluating whether the required technology, tools, and resources are available to successfully develop and implement the proposed system. For this project, the tool request management system is developed using the salesforce platform, which provides built-in features such as experience cloud, apex controllers, lightning web components, flows, reports, and dashboards. These technologies support the development of a web-based system that can efficiently manage student registration, tool requests, and approval processes.

The salesforce developer edition provides a cloud-based environment that allows easy configuration and development without requiring complex hardware or software installation. the platform also supports customization through objects, automation through flows, and communication through email notifications. These features make it technically feasible to develop and implement the entire tool request management process within salesforce.

Since the system is cloud-based, it can be accessed through a web browser and does not require high-end infrastructure or specialized technical equipment. The availability of Salesforce documentation, development tools, and user-friendly configuration options ensures that the system can be developed and maintained efficiently, making the project technically feasible. The platform also provides built-in security features to protect system data and user information.

2.2.2 OPERATIONAL FEASIBILITY

Operational feasibility evaluates whether the proposed tool request management system can be effectively used in the institution. The system is designed to be simple and user-friendly so that students, lab incharge, and head of department can easily perform their tasks without requiring advanced technical knowledge.

Students can register and submit tool requests through the experience cloud portal and the requests are reviewed by the lab incharge and head of department through the approval process. Automated email notifications help inform them about new requests and approvals, making the process faster and more organized. The system helps improve communication between students and laboratory staff. It also ensures proper tracking of tool requests and approvals. Since the system reduces manual work and improves the management of laboratory tools, it is operationally feasible for the institution.

2.2.3 ECONOMIC FEASIBILITY

Economic feasibility evaluates whether the proposed tool request management system is cost-effective and beneficial for the institution. The system is developed using the salesforce developer edition, which does not require additional infrastructure costs. Since the system is cloud-based, there is no need for expensive hardware or software installations. The implementation of this system reduces manual work, paperwork, and time spent managing laboratory tool requests. By automating the request and approval process, the system improves efficiency and reduces operational effort, making it a cost-effective solution for managing laboratory tools.

The system also minimizes the chances of errors that occur in manual record keeping. It helps the institution manage laboratory resources more effectively and maintain accurate records. Maintenance and updates can be handled easily through the salesforce platform. The system improves productivity for both students and laboratory staff. It also supports better monitoring of tool usage and availability. Overall, the system provides long-term benefits with minimal operational cost. The system also ensures faster processing of tool requests. It improves communication between students and laboratory authorities. This helps maintain a more organized and efficient laboratory management process.

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

Software Requirement Specifications (SRS) describe the software requirements needed to develop the tool request management system. It defines the system functions, user interactions, and software components required for proper operation. The system is developed using the salesforce platform with technologies such as Experience Cloud, Apex Controllers, and Lightning Web Components. These technologies help create a web application for student registration, tool request submission, and approval by lab incharge and head of department. The SRS document also explains the system features and operational requirements. It helps developers understand how the system should function during development. It also ensures that the system meets the project objectives and user needs.

2.3.1 HARDWARE REQUIREMENTS

The following hardware requirements are necessary for the development and execution of the proposed tool request management system:

- **Device Type:** Laptop / Desktop Computer
- **Processor:** Intel Core i3 / AMD Ryzen 3 or higher
- **Processor Speed:** 2.0 GHz or higher
- **Installed RAM:** 4 GB minimum (8 GB recommended)
- **System Type:** 64-bit Operating System, x64-based Processor
- **Storage:** Minimum 256 GB Hard Disk / SSD
- **Display Resolution:** 1366 × 768 pixels or higher

2.3.2 SOFTWARE REQUIREMENTS

The following software requirements are necessary for the development and execution of the proposed system:

- **Operating System:** Windows 10 / Windows 11
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Platform:** Salesforce Developer Edition
- **Development Environment:** Salesforce Setup and Experience Builder
- **Technologies Used:** Experience Cloud, Salesforce Flows, Approval Process
- **Reporting Tools:** Salesforce Reports and Dashboards
- **Programming Support:** Salesforce Lightning Platform

2.4 SOFTWARE DESCRIPTION

2.4.1 FRONT END

The front end of the tool request management system is developed using salesforce experience cloud and experience builder, which provides an interactive and user-friendly interface for users to access the system through a web browser. The experience cloud website contains several pages such as home, student registration, tool request, and my tool request, allowing users to easily navigate through different sections of the application. the navigation menu helps users move between pages smoothly, while the interface design ensures that information such as tool details and request forms are clearly displayed. This front-end environment acts as the main platform where students interact with the system.

The front end mainly allows students to register and submit tool requests through the experience cloud portal. Students first complete the student registration form, after which they can submit a tool request form by selecting the required tool and quantity. After the request is submitted, the details are stored in Salesforce and the request is sent for approval. Lab incharge and head of department receive email notifications to review the request and provide approval or rejection. Using experience builder components, forms, and page layouts, the interface is designed to be simple, organized, and easy to use, enabling smooth interaction between users and the salesforce system.

2.4.2 BACK END

The back end of the proposed Tool Request Management System is developed using Salesforce Apex, a strongly typed and object-oriented programming language used to implement server-side logic on the Salesforce platform. Apex connects the Lightning Web Components (LWC) user interface with the Salesforce database and performs operations such as retrieving tool data, processing student requests, validating inputs, and updating records.

The system also uses Salesforce automation tools such as Flows and Approval Processes to handle automated tasks. When a student submits a tool request, email notifications are automatically sent to the lab incharge and head of department for approval. Since the system is hosted on the Salesforce cloud platform, it provides built-in features such as security, scalability, and reliable data management for efficient system operation.

2.4.3 DATABASE

The database of the proposed tool request management system is managed using the salesforce cloud database, which provides a secure and scalable environment for storing application data. It stores information related to student registrations, tool details, and tool request records. The system uses custom objects such as student, tool, and tool request to organize and manage the data. These objects are connected through relationships to maintain data consistency and proper record management. The database operations are performed using SOQL for retrieving records and DML operations in Apex for inserting, updating, and deleting data. Salesforce also provides built-in features such as data backup, authentication, and encryption. This ensures secure storage and efficient management of all system data. The database structure helps in organizing all records in a systematic manner.

It allows quick retrieval of information whenever required by the system. The relationships between objects ensure that tool requests are properly linked with students and tools. The system can also track the status of each request within the database. Salesforce automatically manages database performance and optimization. This reduces the need for manual database maintenance. Overall, the database supports reliable and efficient data management for the entire system.

SUMMARY

This chapter presents the analysis of the proposed tool request management system by identifying the need to improve the manual process used for managing laboratory tool requests in educational institutions. It includes a feasibility study that evaluates the system from technical, operational, and economic perspectives and describes the required software and hardware specifications used for system development. The chapter also explains the limitations of the existing manual system and the need for an automated solution. It identifies the main requirements of the system and how different users interact with it. The analysis helps ensure that the proposed system is efficient and suitable for managing laboratory tool requests. It also provides the foundation for the system design and implementation discussed in the next chapter. The analysis also helps in understanding the system workflow clearly. It ensures that the system meets the needs of students and laboratory authorities.

CHAPTER 3

SYSTEM DESIGN

3.1 MODULE DESCRIPTION

A module description explains the purpose and functionality of each module within a system. It helps in understanding how different parts of the system work together to complete the overall process. In software development, modules divide the system into smaller functional units, making the application easier to develop, manage, and maintain. Each module performs a specific task and interacts with other modules to ensure smooth system operation.

The tool request management system is organized into several modules that manage different stages of the tool request and approval process. These modules work together to provide a complete digital workflow for students, Lab Incharge, and Head of Department. Each module handles a specific function such as student registration, tool request submission, and approval management. The system ensures that tool requests are properly recorded and monitored at every stage. It also helps maintain accurate records of tool usage and request status. The modular structure improves system organization and makes the application easier to update and maintain in the future. This structure also improves system efficiency and reduces processing time. It allows different users to perform their tasks independently within the system. The modular design helps developers easily modify or add new features when required.

The project consists of the following modules:

- Student Registration Module
- Tool Request Module
- My Tool Request Module
- Lab Incharge Approval Module
- Head of Department Approval Module
- Reports and Monitoring Module

3.1.1 STUDENT REGISTRATION MODULE

The tool request module is responsible for managing the process of requesting laboratory tools by students. This module acts as an important part of the system where students can select the required tool and submit their request through the portal. It stores details such as application number, tool name, quantity requested, and course information related to the student's request.

This module allows students to easily submit their tool requests through the experience cloud portal after completing the registration process. The request details are stored in the salesforce system and can be viewed by the lab incharge for further review. The module helps maintain accurate records of all tool requests submitted by students.

The module also ensures that the request is forwarded to the approval process after submission. When a student submits a request, the system sends an email notification to the lab incharge for verification. If approved, the request is then forwarded to the Head of Department for final approval. By maintaining proper request information and approval tracking, this module ensures efficient management of laboratory tool requests.

3.1.2 TOOL REQUEST MODULE

The tool request module is responsible for allowing students to view and track the tool requests they have submitted through the system. This module helps students check the details and status of their requests, ensuring transparency in the tool request process. It displays information such as application number, requested tool name, quantity, request date, and approval status. In this module, students can access their submitted requests through the experience cloud portal. The system displays request details and shows whether the request is pending, approved, or rejected. The request status is automatically updated when the lab incharge or head of department reviews the request. This helps students easily track their tool requests without contacting laboratory staff. The module also helps students verify the information they submitted in the request form. It ensures that all request details are clearly visible to the user. This improves user convenience and system transparency.

3.1.3 MY TOOL REQUEST MODULE

The tool request module is responsible for allowing students to view and track the tool requests they have submitted in the system. This module helps students check the status of their requests and view the details related to the requested tools. It provides transparency in the tool request process by showing whether a request is pending, approved, or rejected.

In this module, students can access their request details through the Experience Cloud portal using their Application Number. The system retrieves the stored request records and displays information such as tool name, quantity requested, request date, and approval status. This allows students to easily monitor the progress of their requests.

The system automatically updates the request status when the lab incharge or head of department reviews the request. Students can view the updated status without contacting laboratory staff. This module improves user convenience and helps maintain proper tracking of all tool requests submitted by students.

3.1.4 LAB INCHARGE APPROVAL MODULE

The lab incharge approval module is responsible for reviewing and managing the tool requests submitted by students. This module allows the lab incharge to verify the details of each request and check the availability of the requested laboratory tools before taking any action.

In this module, the lab incharge can view the list of tool requests submitted by students along with details such as application number, tool name, quantity requested, and request date. After reviewing the request, the lab incharge can approve or reject the request based on tool availability and laboratory requirements.

If the lab incharge approves the request, the system automatically forwards it to the head of department for final approval. An email notification is sent to the head of department to review the request. This module ensures proper verification of tool requests and helps maintain effective management of laboratory tools.

3.1.5 HEAD OF DEPARTMENT APPROVAL MODULE

The head of department approval module is responsible for providing the final approval for tool requests submitted by students. After the lab incharge approves a tool request, the request is automatically forwarded to the head of department for further review. This module ensures that all tool requests are verified at a higher level before final approval.

In this module, the head of department can view the details of the request such as application number, tool name, quantity requested, and approval status from the lab incharge. The head of department reviews the request and decides whether to approve or reject it based on the laboratory requirements and availability of resources.

Once the head of department makes a decision, the system updates the final status of the tool request in the database. This module ensures proper authorization, improves transparency, and helps maintain effective control over the usage of laboratory tools.

3.1.6 REPORTS AND MONITORING MODULE

The reports and monitoring module is responsible for analyzing and tracking the tool request data in the system. This module helps administrators and department authorities monitor the usage of laboratory tools and review the requests submitted by students. It provides a clear overview of all tool requests and their approval status. In this module, users can view reports related to student registrations, tool requests, approved requests, and rejected requests. These reports help authorities understand how laboratory tools are being used and identify request trends within the institution.

The module uses salesforce reports and dashboards to present the data in an organized and easy-to-understand format. This helps improve decision-making and ensures better monitoring and management of laboratory tools. The reports can be generated quickly whenever required by the administrators. The data visualization helps users easily interpret system information. It also supports better tracking of tool availability and request history. Overall, the module improves transparency and efficient management of laboratory resources.

3.2 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that shows the flow of data through a system. It illustrates how data moves from one entity to another, how it is processed, and how it is stored within the system. DFDs help in understanding the overall functionality of the tool request management system and the interaction between external users and internal processes. it also helps identify the inputs and outputs involved in different processes such as student registration, tool request submission, and approval by lab incharge and head of department. DFDs provide a clear visual representation that makes complex system operations easier to understand and analyze.

3.2.1 DFD LEVEL 0

A Data Flow Diagram (DFD) Level 0, also known as a context diagram, represents the overall view of the tool request management system. It shows the entire system as a single process and illustrates how external entities such as students, lab incharge, and head of department interact with the system. The diagram focuses on the flow of data between the system and these external users without showing the detailed internal processes. DFD Level 0 helps in understanding the basic functionality of the system and identifies the main inputs and outputs

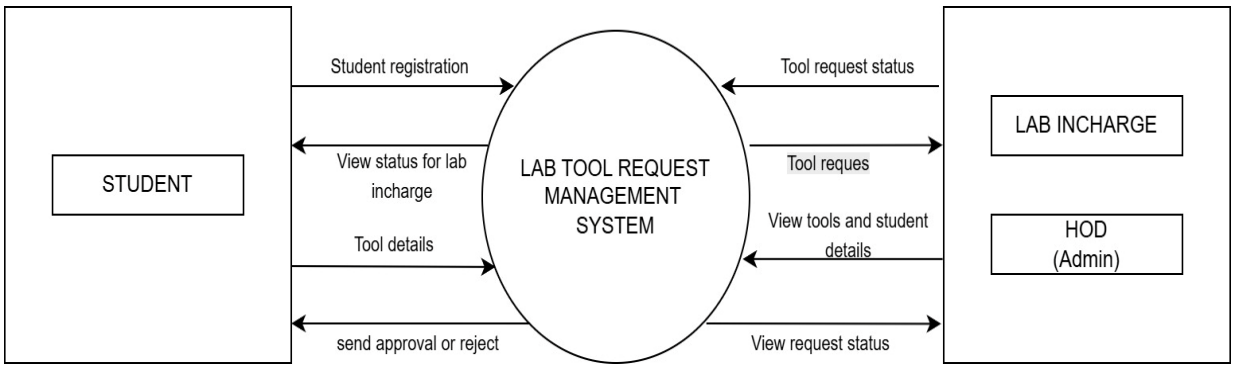


Figure 3.1 Data Flow Diagram Level 0

Figure 3.1, the DFD Level 0 represents the overall view of the lab tool request management system and shows how different external users interact with the system. In this diagram, the entire system is represented as a single process, and the interactions between the system and external entities such as student, lab incharge, and head of department are shown through data flows.

In the system, students can register, view tool details, and submit tool requests through the portal. They can also check the status of their requests after submission. The system receives the request details from students and stores them in the database for further processing. Students can also view whether their request has been approved or rejected.

The Lab Incharge interacts with the system to view tool requests submitted by students and review the request details. After verifying the request and checking tool availability, the Lab Incharge can approve or reject the request. If the request is approved, the system forwards it to the Head of Department (HOD) for final approval.

The head of department (admin) can access the system to review the tool requests approved by the Lab Incharge and make the final decision to approve or reject them. The system then updates the request status and sends the result back to the student. The DFD Level 0 diagram helps in understanding the overall flow of information between users and the system, showing how data moves between external entities and the system during the tool request and approval process.

3.2.2 DFD LEVEL 1

DFD Level 1 provides a detailed representation of the lab tool request management system by breaking down the main process shown in DFD level 0 into smaller sub-processes. It shows how different processes within the system handle and process data. The diagram also illustrates the flow of information between external entities, processes, and data stores. Each process performs a specific function within the system. DFD level 1 helps in understanding how data moves step by step inside the system and provides a clearer view of the internal working of the system.

In this level, the system functions are divided into processes such as student registration, tool request submission, lab incharge approval, head of department approval, and tool issue management. It also shows how different users such as student, lab incharge, and

head of department interact with these processes. The diagram indicates where the data is stored, such as Student records, Tool details, and Tool Request records, and how the data is retrieved when needed. This helps in identifying the relationship between processes and data stores. Overall, DFD Level 1 provides a more detailed understanding of the operations of the system.

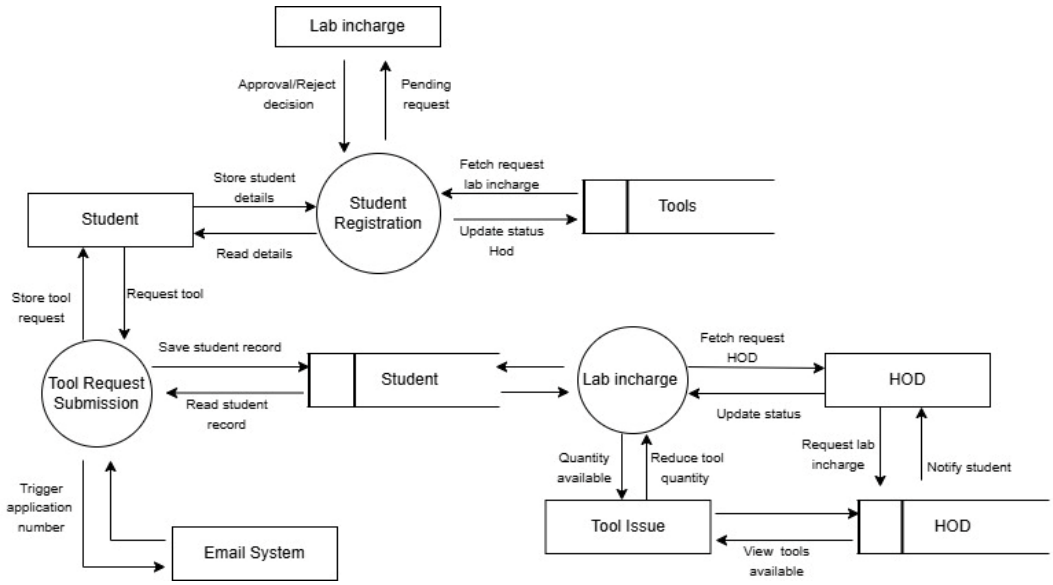


Figure 3.2 Data Flow Diagram Level 1

Figure 3.2, the DFD Level 1 diagram provides a detailed view of the Lab Tool Request Management System by breaking down the main system into smaller processes and showing how data flows between users, processes, and data stores. The main users involved in this system are student, lab incharge, and head of department. In the system, the student first completes the student registration process, where the student details are stored in the student database. After registration, the student can submit a request for laboratory tools through the tool request submission process. The system records the request details and sends an email notification to the Lab Incharge about the new tool request.

The lab incharge reviews the submitted tool requests and checks the request details and tool availability in the data store. Based on the verification, the lab incharge can approve or reject the request. If approved, the request is forwarded to the head of department for final approval. The head of department reviews the request and makes the final decision. After approval, the system updates the request status and tool availability. The student can view the updated request status through the system.

3.3 USECASE DIAGRAM

A use case diagram is a graphical representation used in system design to show the interaction between users and the system. It illustrates the different functions or services provided by the system. The users who interact with the system are called actors, and the actions or operations performed by these actors are known as use cases. Use case diagrams help in clearly understanding the system requirements and how different users interact with the system features. They also represent the relationship between actors and system functions. These diagrams are mainly used during the system analysis and design phase and provide a clear overview of how the system operates.

In the Lab Tool Request Management System, the use case diagram includes actors such as Student, Lab Incharge, and Head of Department. Each actor interacts with the system in different ways according to their responsibilities. The student can perform actions such as student registration, submit tool request, and view request status. These functions allow students to register in the system, request laboratory tools, and check the approval status of their requests.

The lab incharge is responsible for reviewing the tool requests submitted by students. the lab incharge interacts with the system to perform tasks such as tool approval or rejection and lab incharge approval. The lab incharge verifies the request details and checks the availability of tools before approving or rejecting the request. This ensures that the requested tools are issued only when they are available and necessary.

The head of department performs higher-level administrative tasks in the system. The head of department can Approve or Reject tool requests, manage system records, and monitor the tool issuing process. After reviewing the request forwarded by the Lab Incharge, the head of department provides the final decision regarding the tool request. Once approved, the system allows the tool to be issued and the request status is updated. The head of department also ensures that tools are used properly for academic purposes. This helps maintain control, transparency, and efficient management of laboratory tools in the system. The head of department can also review the history of tool requests. This helps in monitoring the usage of laboratory tools. It also supports better management and decision-making.

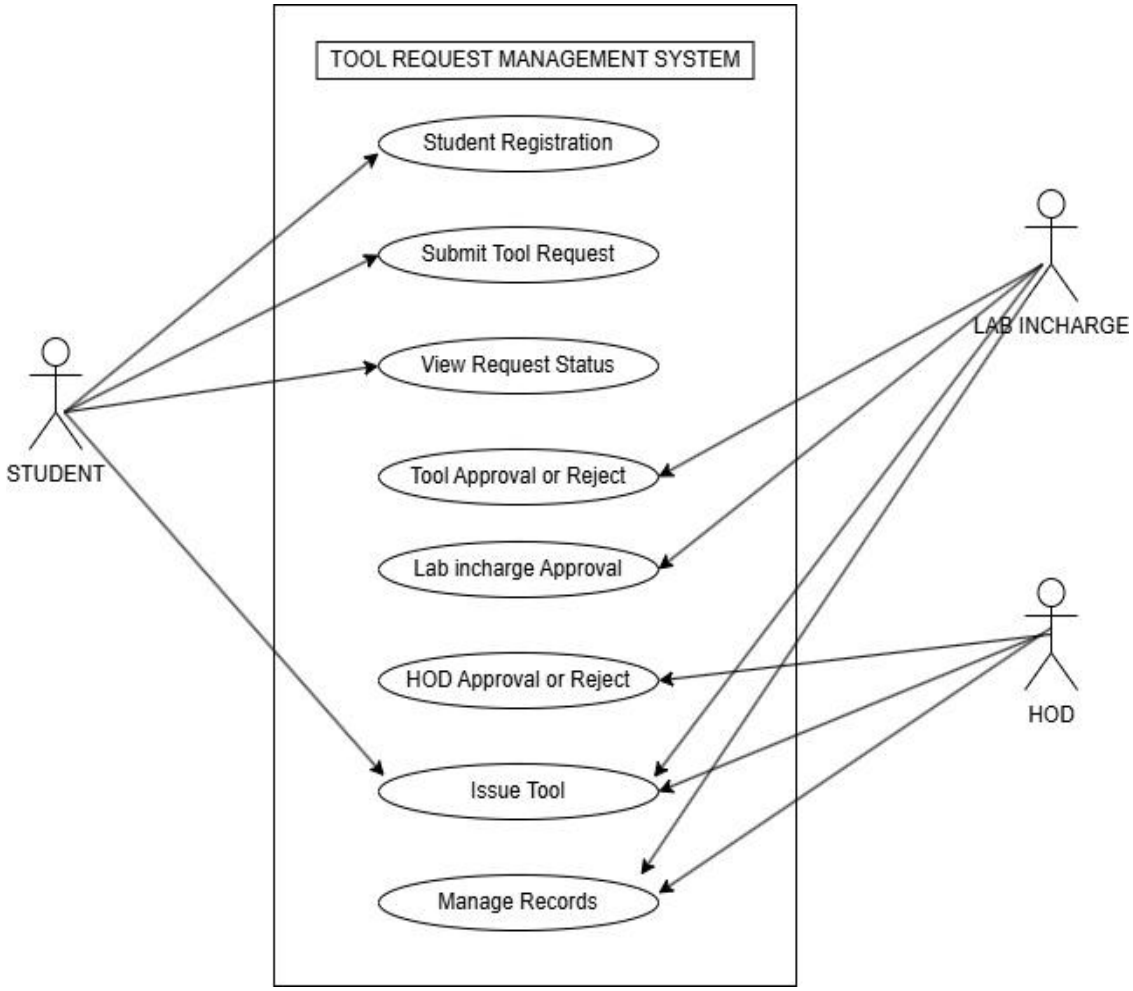


Figure 3.3 Use Case Diagram

Figure 3.3, the Use Case Diagram of the lab tool request management system represents the interaction between different users and the system functions. It shows how the main actors such as student, lab incharge, and head of department interact with the system to perform various activities. The diagram helps in understanding the responsibilities of each user and the processes involved in managing laboratory tool requests. It clearly illustrates how students submit requests, how the Lab Incharge reviews them, and how the head of department provides final approval before tools are issued.

It also shows the flow of actions from request submission to final approval. The diagram helps in visualizing the overall working of the system clearly. It ensures that each user role and system function is well organized and properly defined. This improves the efficiency and management of laboratory tool requests within the system.

3.4 INPUT DESIGN

Input design involves creating user interfaces that allow users to enter data into the system accurately and efficiently. A well-designed input interface reduces errors, improves usability, and ensures smooth interaction between the user and the system. Proper input design focuses on simplicity, clarity, validation, and ease of use so that users can easily provide the required information.

In the lab tool request management system, input forms are designed to allow users such as students, lab incharge, and head of department to enter and manage information effectively. The system includes forms for student registration and tool request submission, where students provide details such as name, email, application number, tool name, and quantity. These input forms ensure that data is entered correctly and stored securely in the system.

The input design also includes validation features that help prevent incorrect or incomplete data from being submitted. Required fields ensure that important information is not left blank, while appropriate data types help maintain data accuracy, such as validating email format and numeric values. These features improve the reliability of the system and help maintain accurate records. Overall, effective input design supports the smooth functioning of the lab tool request management system.

The input forms are designed with a simple and user-friendly layout to make it easy for users to enter information. Clear labels and instructions are provided to guide users while filling the forms. The system also displays error messages if incorrect or incomplete data is entered. This helps users correct mistakes before submitting the form. The interface is designed to ensure that users can complete tasks quickly without confusion. Proper input design also helps reduce the chances of duplicate or invalid records. It ensures that the data stored in the system remains accurate and consistent. The organized input structure helps administrators manage the records efficiently. This also improves the overall performance and reliability of the system. The forms are designed to support smooth data entry for all users. It also improves the overall user experience of the system. This helps maintain better data management and system efficiency. The structured layout makes the forms easy to understand. It also ensures that users can submit requests without difficulty.

STUDENT SEARCH

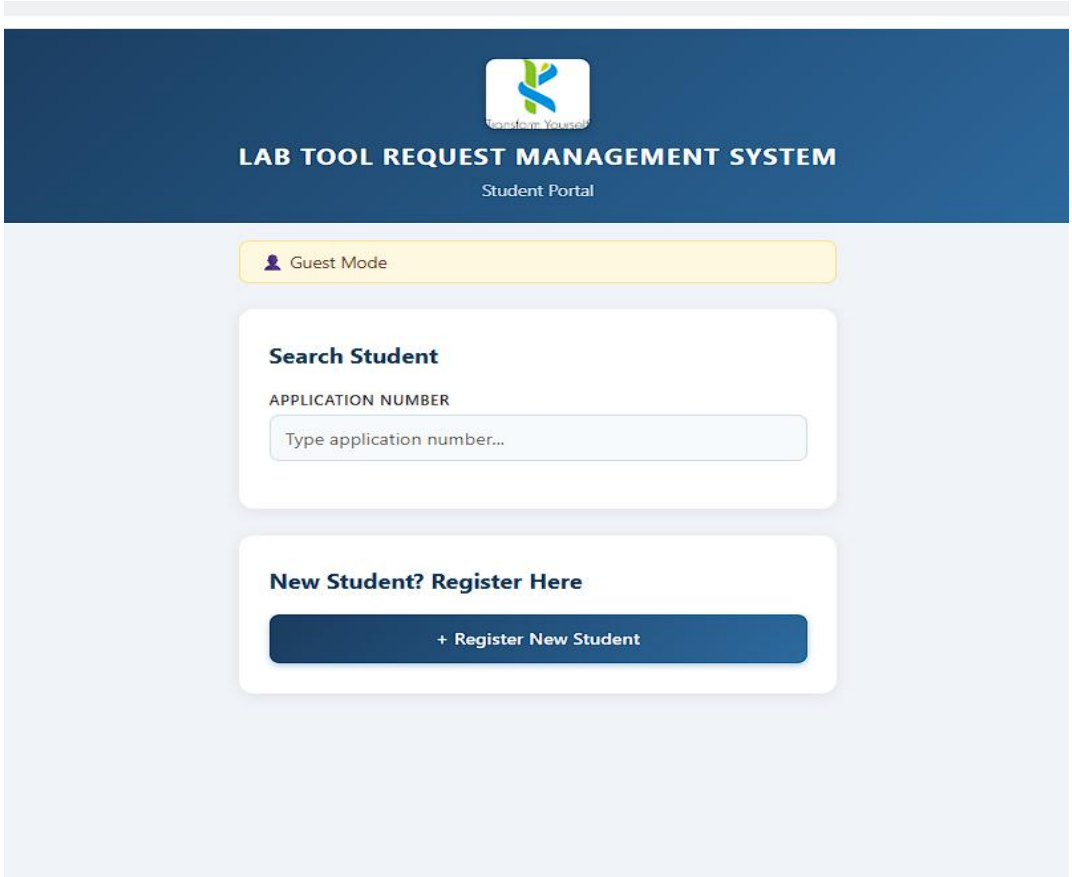


Figure 3.4 Student Search

Figure 3.4, the Student Search page is used to identify registered students in the lab tool request management system before submitting a tool request. It requires students to enter their application number to search for their registration details in the system. This page ensures that only registered students can access the tool request features. After entering the correct application number, the student can proceed to submit a tool request.

This input screen helps the system retrieve the correct student information stored in the salesforce database. if the application number is valid, the system allows the student to continue to the tool request form. if the application number is incorrect, the system prompts the user to check the entered details. The student search page also helps maintain proper identification of students in the system. It ensures that tool requests are linked to the correct student records. This makes it easier for the lab incharge and head of department to review requests and manage laboratory tool usage efficiently.

STUDENT REGISTRATION

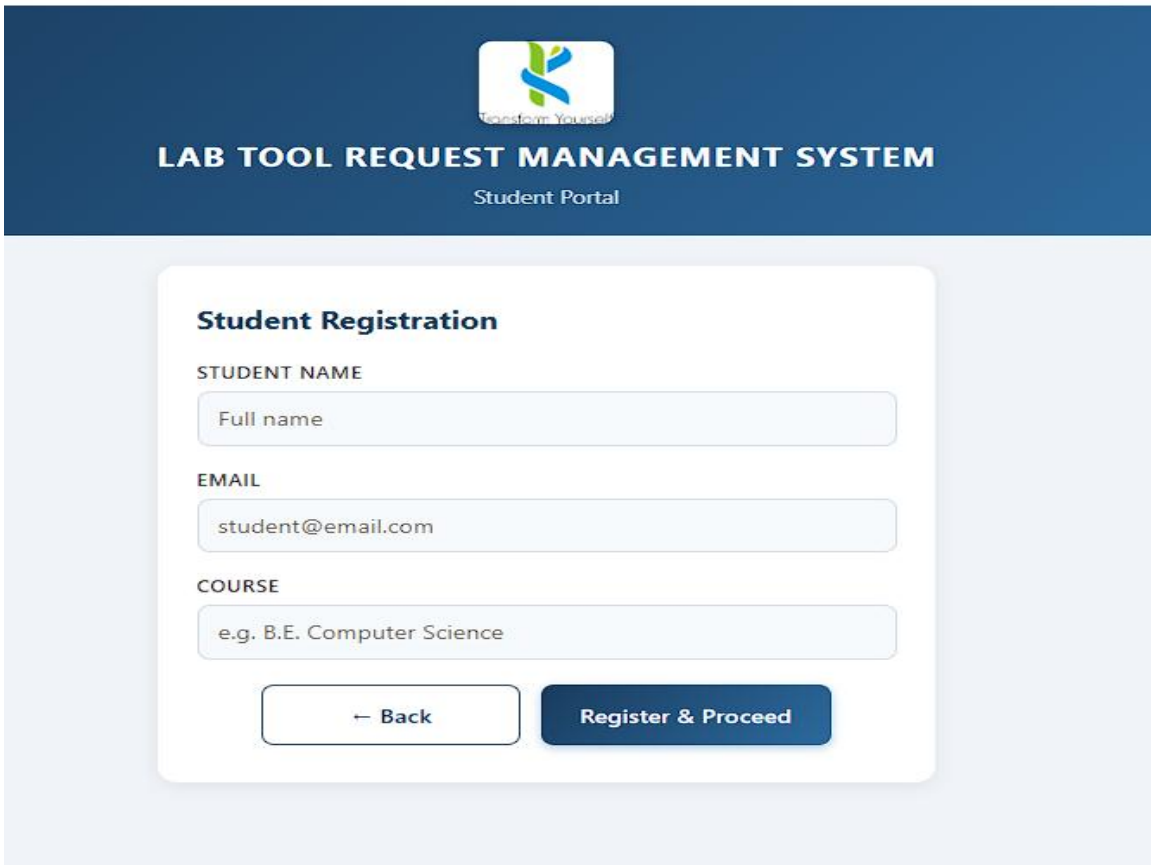


Figure 3.5 Student Registration

Figure 3.5, the Student Registration page is used to collect student details before accessing the lab tool request management system. Students enter information such as Student Name, Email, and Course to register in the system. After entering the details, the student clicks the register & proceed button to complete the registration. Once the registration is successful, the system generates a unique application number for the student.

This input screen ensures that student details are entered correctly and stored securely in the salesforce database. validation features help ensure that required fields are filled correctly before submission. This helps prevent incorrect or incomplete data from being stored in the system. The registration page also helps maintain accurate student records in the system. The stored information can be accessed by the lab incharge and head of department when reviewing tool requests. After successful registration, students can proceed to the tool request form to request laboratory tools.

TOOL REQUEST

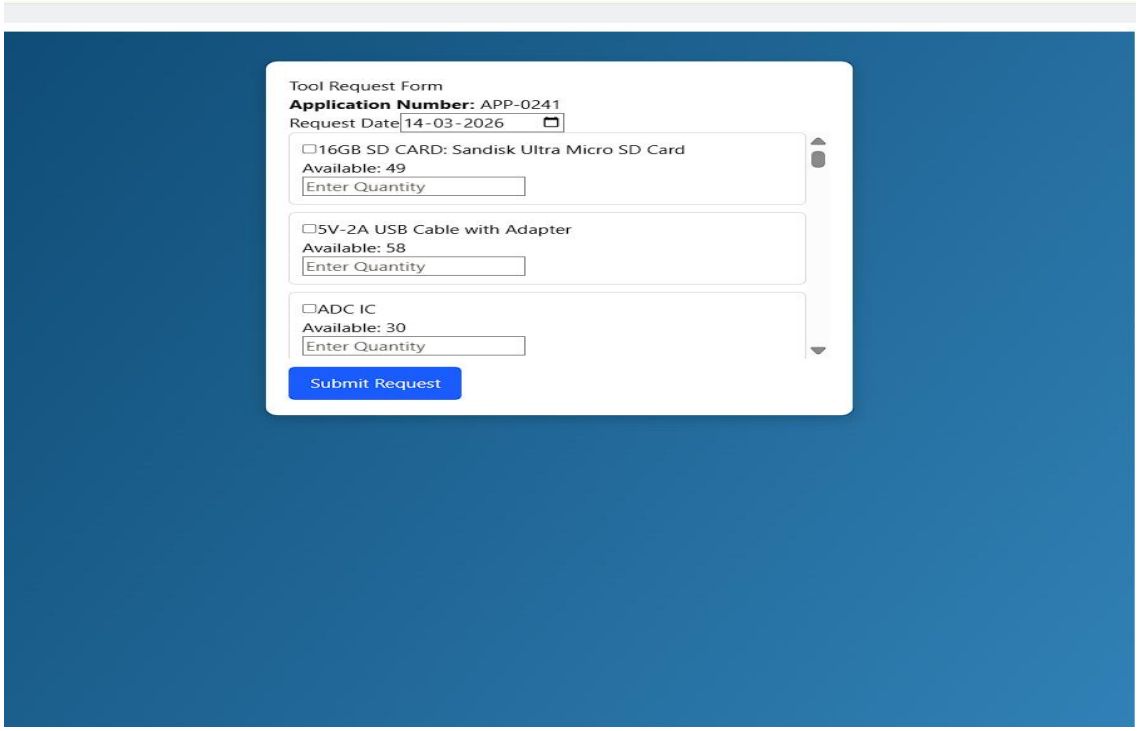


Figure 3.6 Tool Request

Figure 3.6, this input screen is used by the student to submit a tool request in the lab tool request management system. The form allows the student to enter details such as application number, select the required tool, and specify the quantity needed. These details help the system record the tool request and link it with the registered student in the database. After entering the required information, the student can submit the request using the submit button.

Once the request is submitted, the system stores the request details in the salesforce database and sends an email notification to the lab incharge for review. The lab incharge verifies the request and decides whether to approve or reject it. if the request is approved, it is forwarded to the head of department for final approval.

This input form helps ensure that tool requests are recorded accurately and managed in an organized manner. It also allows the system to track the request status and maintain proper records of tool usage. The stored information can be viewed later through the My Tool Request page, allowing students to monitor the progress of their requests.

STAFF LOGIN

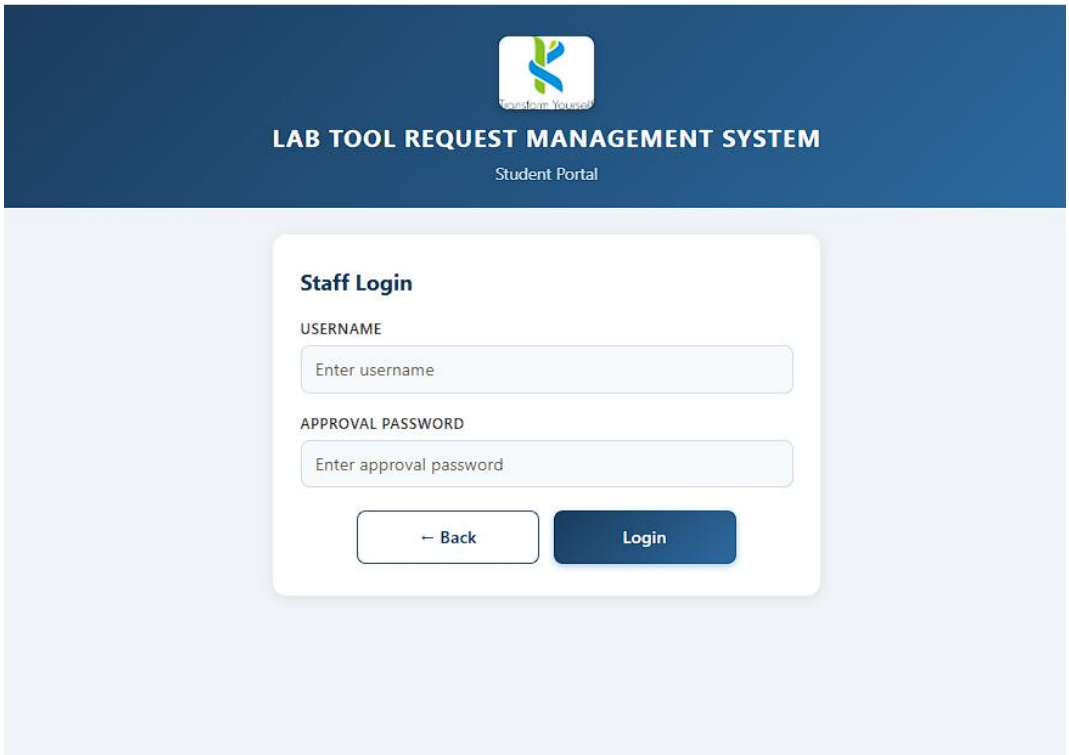


Figure 3.7 Staff Login

Figure 3.7, the Staff Login page is used to authenticate authorized staff members before accessing the lab tool request management system. It requires staff such as lab incharge and head of department to enter their username and approval password to log into the system. This page ensures that only authorized personnel can access the approval and management features of the system. After entering the correct credentials, the user can click the login button to access the staff dashboard.

This input screen helps maintain system security and controlled access to the application. It prevents unauthorized users from accessing sensitive information related to tool requests and approvals. The login page verifies the credentials entered by the staff and allows access only if the information is valid. The staff login page also helps the system identify the user's role and permissions within the system. Once the staff member successfully logs in, they can review tool requests submitted by students and approve or reject them accordingly. This ensures that the tool request approval process is handled securely and efficiently within the lab tool request management system.

TOOL REQUEST HISTROY

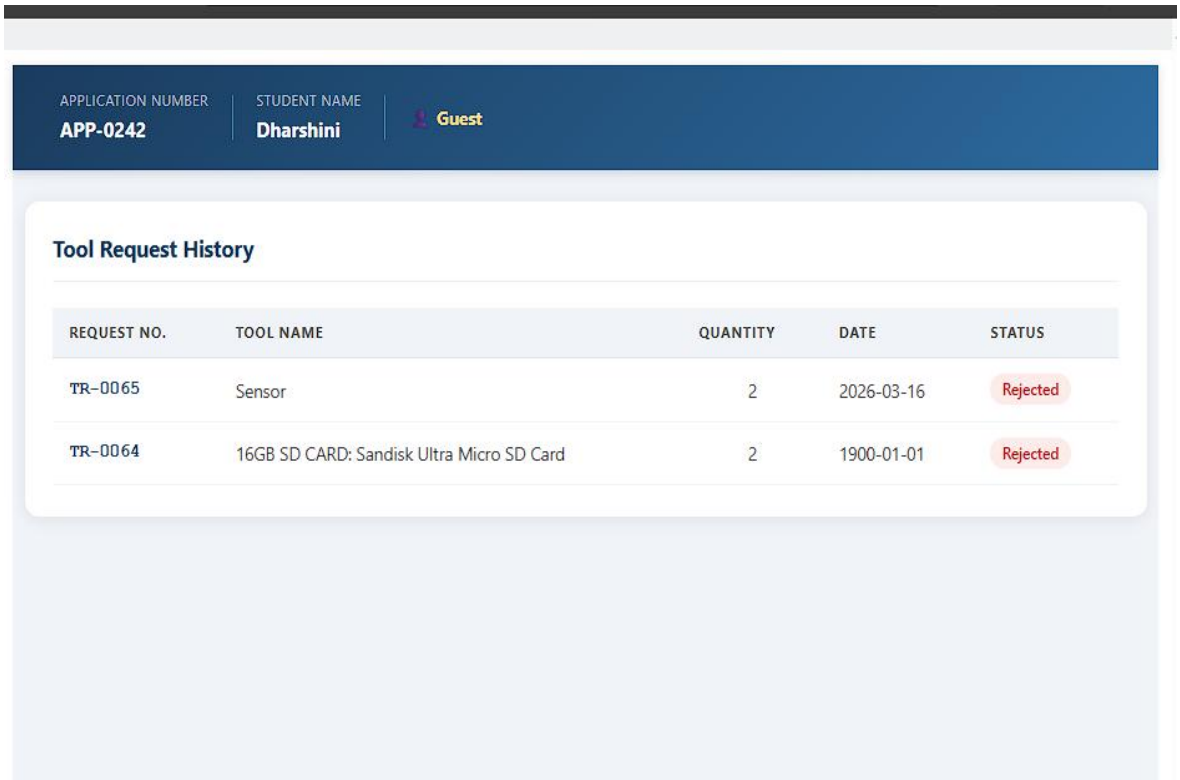


Figure 3.8 Tool Request History

Figure 3.8, this input screen is used to view the tool request history in the lab tool request management system. It allows students to check the details of the tool requests they have previously submitted. The screen displays information such as Application Number, Tool Name, Quantity Requested, Request Date, and Request Status. This helps students easily track the progress of their requests in the system.

The tool request history page retrieves the request details stored in the salesforce database and displays them in an organized format. Students can view whether their request is Pending, Approved, or Rejected. This allows them to stay informed about the status of their tool requests without contacting laboratory staff.

This screen helps maintain transparency in the system by allowing students to monitor their submitted requests. It also ensures that all tool request records are stored properly and can be accessed whenever needed. The tool request history page improves user convenience and supports effective management of tool requests in the system.

3.5 OUTPUT DESIGN

STUDENT DETAILS

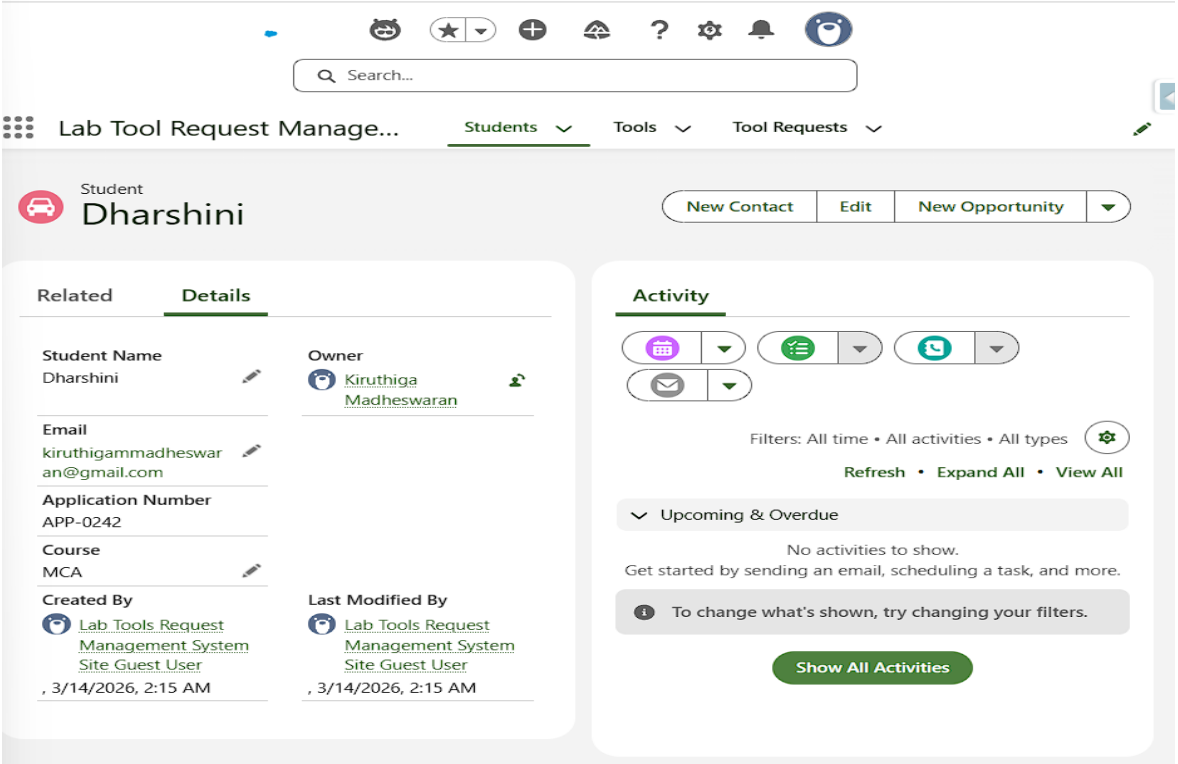


Figure 3.9 Student Details

Figure 3.9, this output screen displays the student details record in the lab tool request management system. it shows the stored information of a registered student such as student name, email, application number, and course. These details help maintain accurate student records in the system. The screen also shows additional information such as the record owner, created date, and last modified details, which helps track the record management within the system.

The output is organized into sections such as details and activity, allowing users to easily view the stored information and related activities. The details section displays the main student information, while the activity section shows actions such as emails, tasks, or other updates related to the student record. This structured layout helps administrators quickly access important information. this output screen helps lab incharge and head of department verify student details when reviewing tool requests.

TOOL REQUEST RECORD

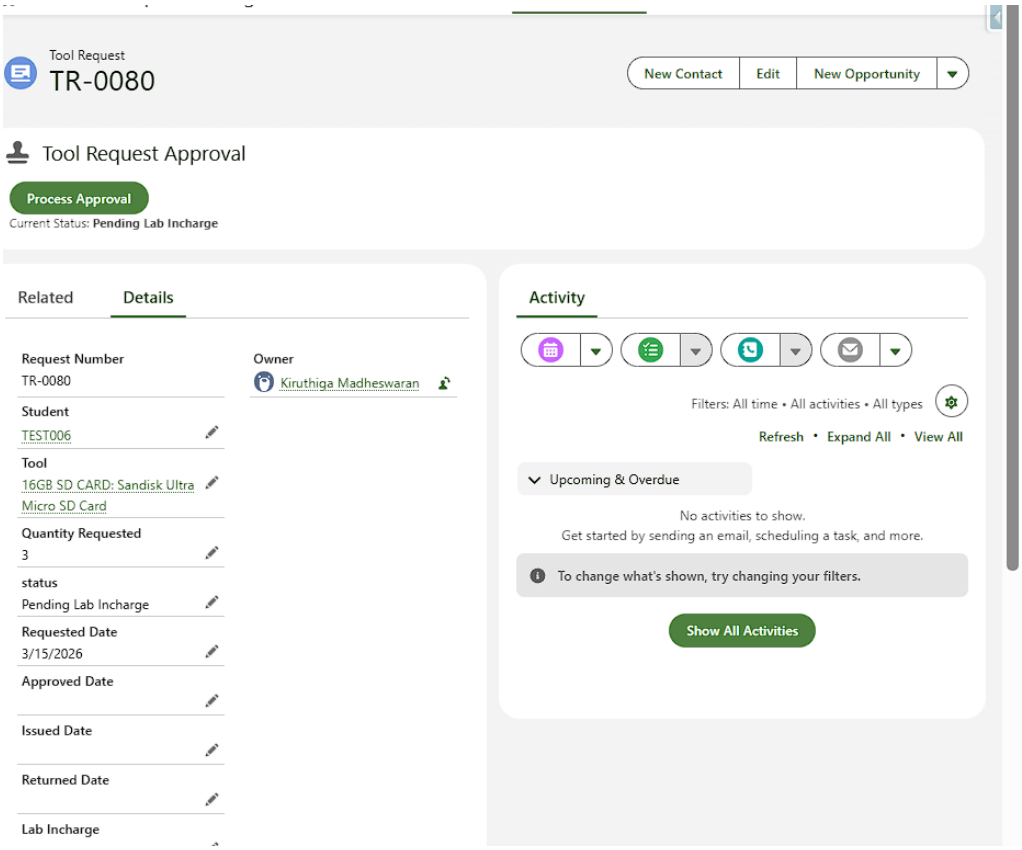


Figure 3.10 Tool Request Record

Figure 3.10, this output screen displays the tool request record in the lab tool request management system. It shows the details of a tool request submitted by a student, including information such as Request Number, Student Name, Tool Name, Quantity Requested, Request Status, and Requested Date. These details help maintain accurate records of tool requests in the system.

The output is organized within the details section, where users can view the complete information related to the tool request. Each request is assigned a unique request number, which helps identify and track the request easily. The screen also displays the approval status, allowing users to know whether the request is pending, approved, or rejected. This output helps the lab incharge and head of department review and process tool requests efficiently. It ensures that all request information is clearly displayed and can be accessed whenever required. Overall, this output design improves data visibility, request tracking, and efficient management of laboratory tools within the system.

LAB INCHARGE APPROVAL

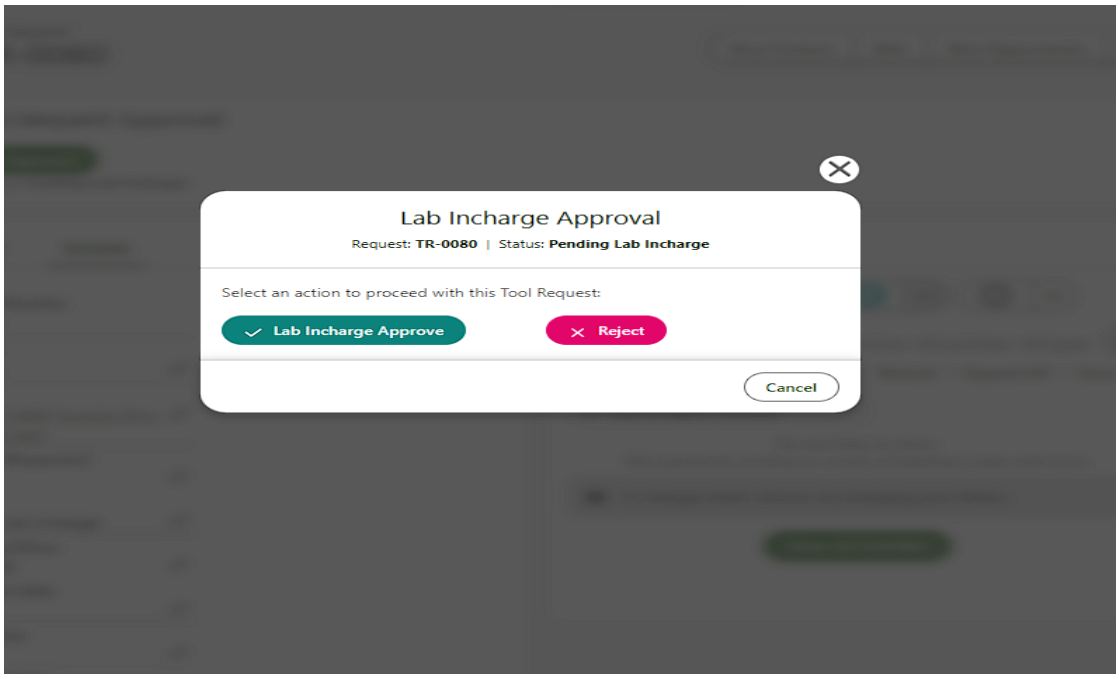


Figure 3.11 Lab Incharge Approval

Figure 3.11, this output screen displays the Tool Request Approval page in the lab tool request management system. It shows the details of a tool request submitted by a student and allows the Lab Incharge to review the request before taking action. The screen displays information such as request number, student name, tool name, quantity requested, request status, and requested date. These details help the lab incharge verify the request and check the availability of tools in the laboratory.

The output is organized in the details section, where all request information is clearly displayed. The screen also includes the process approval option, which allows the lab incharge to proceed with the approval process. When the lab incharge selects this option, the system displays actions such as approve or reject the tool request.

This output screen helps the lab incharge manage and control tool requests efficiently. It ensures that every request is reviewed before approval and that the request status is updated correctly in the system. Overall, this output design improves request monitoring, approval management, and proper utilization of laboratory tools within the system.

TOOL REQUEST STATUS-REJECTED

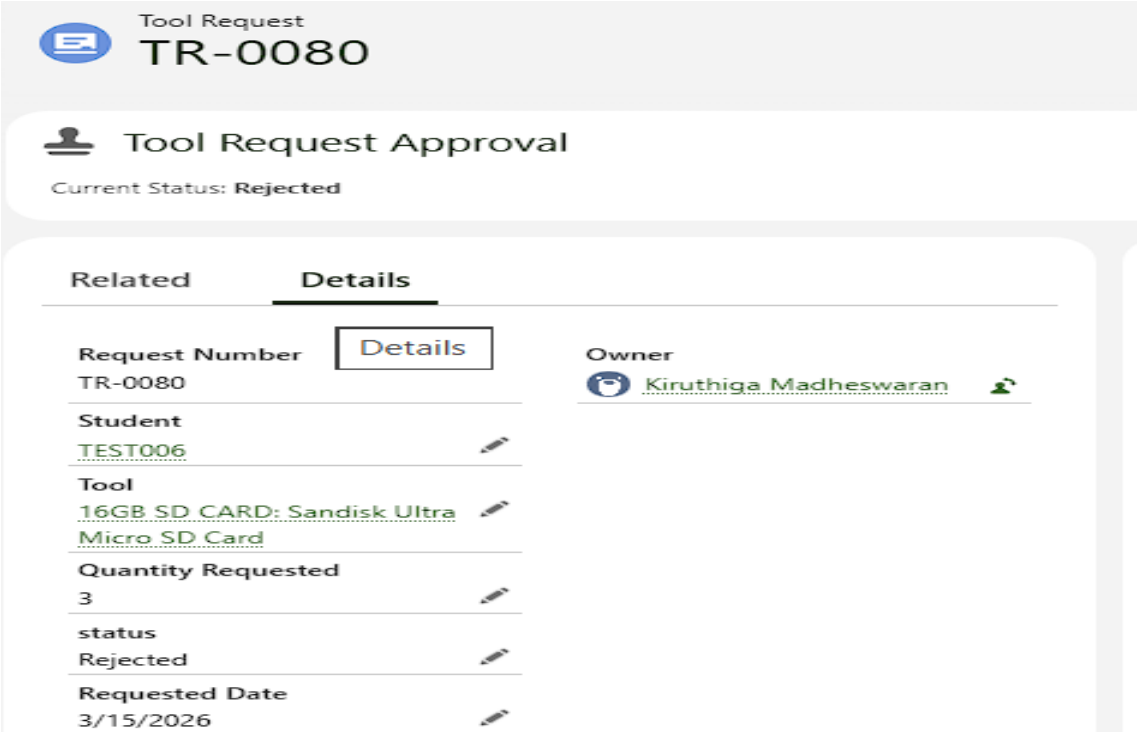


Figure 3.12 Tool Request Status-Rejected

Figure 3.12, this output screen displays the rejected tool request details in the lab tool request management system. It shows information related to the tool request such as request number, student name, tool name, quantity requested, requested date, and status. The status field clearly indicates that the tool request has been rejected by the lab incharge or head of department.

The output is organized in the details section, where users can easily view all the information related to the selected request. Each request is identified with a unique request number, which helps users quickly locate and review the request record stored in the system. This layout ensures that all request details are displayed in a clear and structured format.

This output helps the lab incharge and head of department verify the decision taken on a tool request and maintain proper records of rejected requests. It also allows administrators to track request history and monitor system activities effectively. Overall, this output design improves data visibility, request tracking, and proper management of tool request records within the system.

TOOL REQUEST STATUS APPROVED

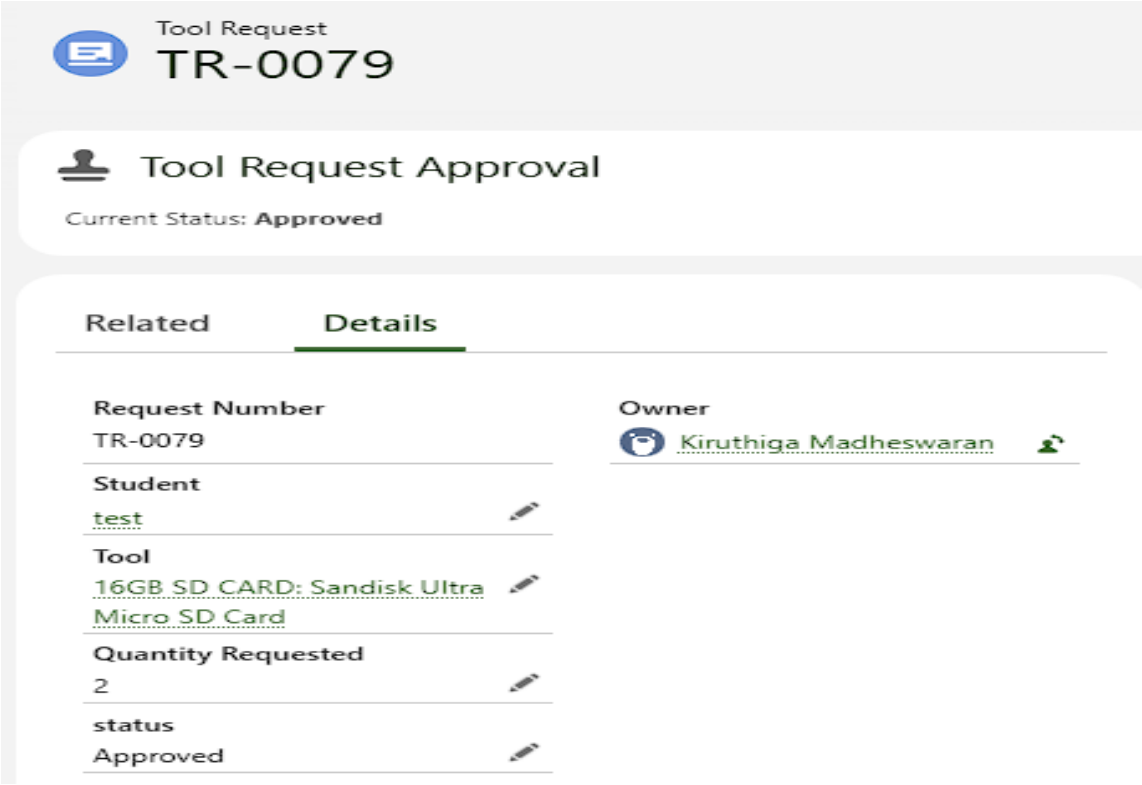
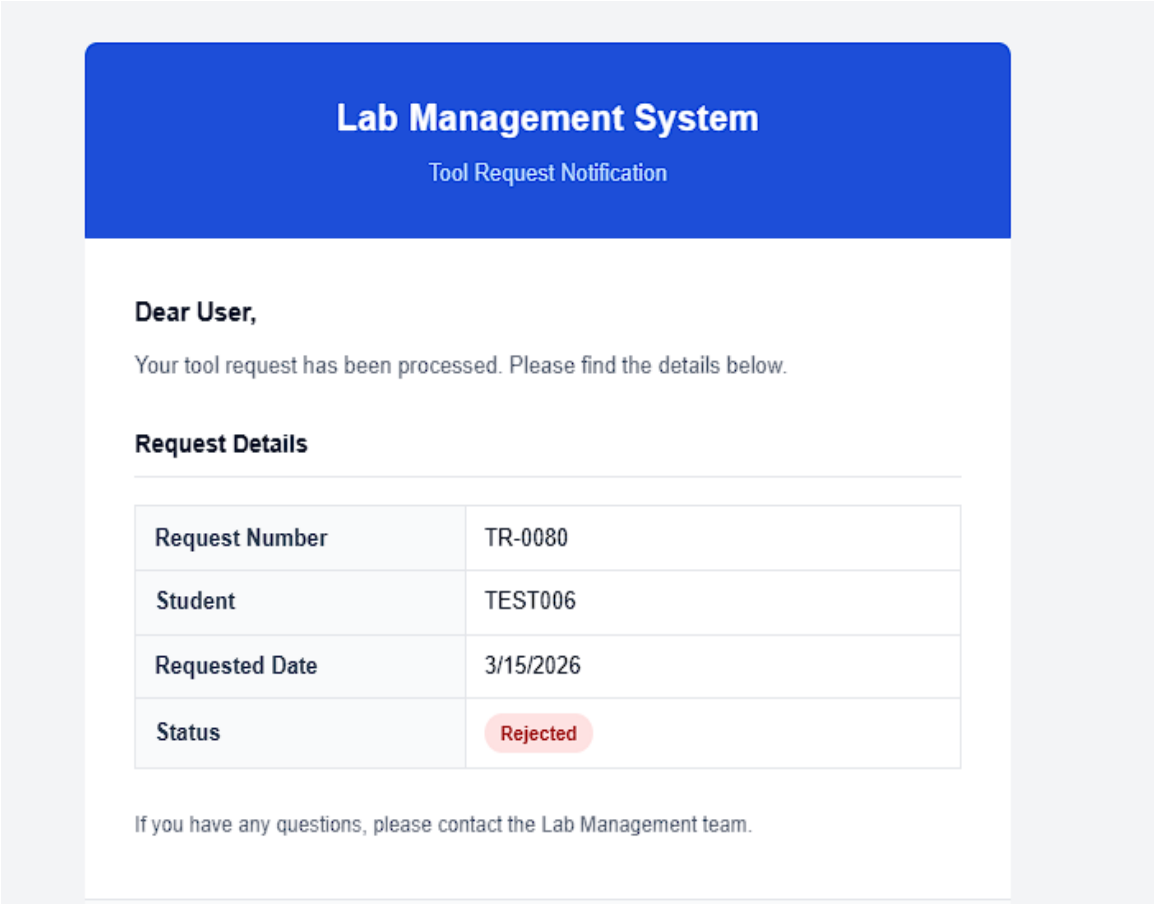


Figure 3.13 Tool Request Status-Approved

Figure 3.13, this output screen displays the approved tool request details in the lab tool request management system. It shows the information related to a specific tool request submitted by a student. The screen displays details such as request number, student name, tool name, quantity requested, request status, requested date, and approved date. These details help users clearly view the status and information of the approved tool request.

The output is organized in the details section, where all request information is displayed in a structured format. The request status field shows “Approved” indicating that the tool request has been successfully approved by the lab incharge or head of department. This layout allows administrators to easily review the request details and verify the approval status. This output helps lab incharge and head of department monitor the approval process and track tool requests efficiently. It ensures that the approved request records are stored properly and can be accessed whenever required. Overall, this output design improves request tracking, transparency, and effective management of laboratory tools in the system.

EMAIL NOTIFICATION REJECTED



Lab Management System

Tool Request Notification

Dear User,

Your tool request has been processed. Please find the details below.

Request Details

Request Number	TR-0080
Student	TEST006
Requested Date	3/15/2026
Status	Rejected

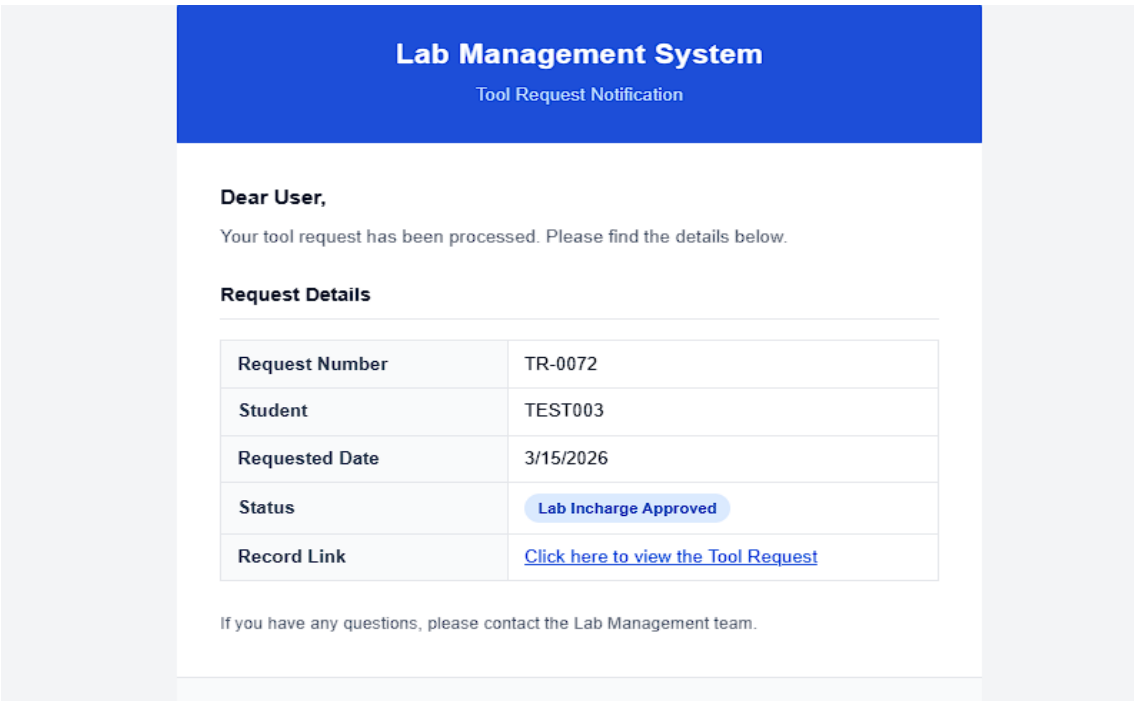
If you have any questions, please contact the Lab Management team.

Figure 3.14 Email Notification Rejected

Figure 3.14, this output screen displays the email notification sent when a tool request is rejected in the lab tool request management system. It shows important details such as request number, student name, requested date, and request status, indicating that the request has been rejected. The email is automatically generated by the system after the rejection decision is made. This notification helps inform the student about the request outcome.

The email is displayed in a clear and structured format, making it easy to read the request information. It ensures that students receive immediate updates about their tool request status. This output also helps maintain proper communication between the system and the users. Overall, the email notification improves transparency and efficiency in the tool request approval process.

APPROVAL EMAIL NOTIFICATION



Lab Management System

Tool Request Notification

Dear User,

Your tool request has been processed. Please find the details below.

Request Details

Request Number	TR-0072
Student	TEST003
Requested Date	3/15/2026
Status	Lab Incharge Approved
Record Link	Click here to view the Tool Request

If you have any questions, please contact the Lab Management team.

Figure 3.15 Email Notification Approval

Figure 3.15, this output screen displays the email notification sent after a tool request is approved in the lab tool request management system. The email contains important information about the request, including details such as request number, student name, requested date, and request status. This notification helps inform the concerned user that the tool request has been processed successfully.

The email output is organized in a clear and structured format, where the request details are displayed in a table for easy readability. It also includes a record link that allows users to directly view the tool request in the system. This helps users quickly access the request information and check additional details if needed.

This output helps lab incharge, head of department, and students stay updated about the approval status of the tool request. It ensures that the request processing information is communicated clearly through automated notifications. Overall, this output design improves communication, transparency, and efficient management of tool request approvals within the system.

3.6 ER DIAGRAM

An ER Diagram is a graphical representation used to show the structure of a database. It illustrates the entities present in a system and the relationships between them. Entities represent real-world objects such as Student, tool request, lab incharge, tool, and head of department in the lab tool request management system. The diagram also shows attributes that describe each entity. ER diagrams help in designing and organizing the database structure of a system.

It also helps developers understand how different data elements are connected within the system. The diagram clearly shows how data flows between entities and how relationships are maintained. Each entity contains attributes that store specific information related to that entity. These relationships help ensure proper data organization and reduce data redundancy in the database. Overall, the ER diagram provides a clear structure for storing and managing system data efficiently.

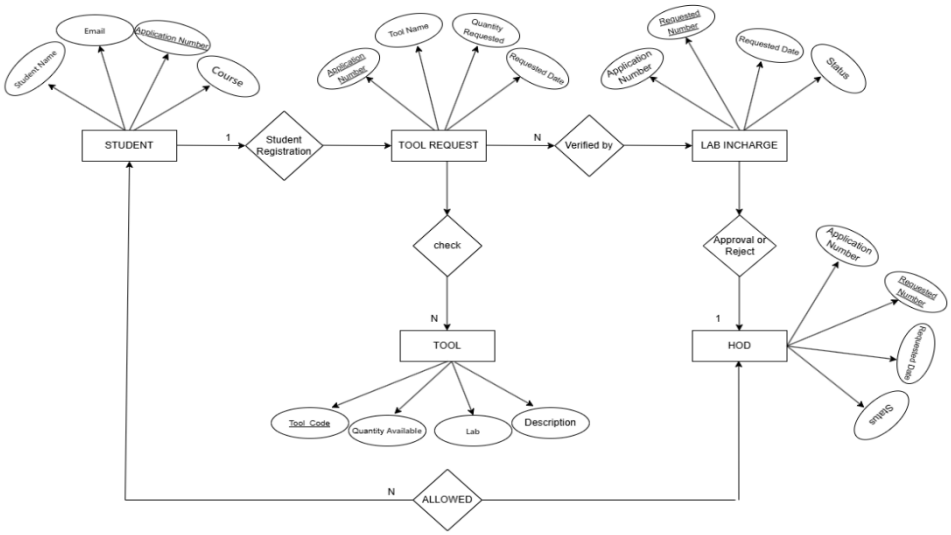


Figure 3.16 ER DIAGRAM

Figure 3.16, the ER diagram of the lab tool request management system shows the relationship between entities such as student, tool request, tool, lab incharge, and head of department. It helps organize data related to tool requests and approvals. The student submits tool requests, the lab incharge reviews them, and the head of department gives final approval. This structure helps manage tool request information efficiently.

3.7 DATABASE DESIGN

Database design refers to the process of organizing and structuring data so that it can be stored, managed, and retrieved efficiently. It defines how data is stored in the system and how different data elements are related to each other. A well-designed database ensures data accuracy, consistency, and easy access to information. Proper database design also helps reduce data redundancy and improves the overall performance of the system.

In the lab tool request management system, the database is implemented using the salesforce platform, where data is stored in the form of objects, fields, and relationships. Objects are similar to database tables and store records related to different entities in the system, while fields store specific details such as student name, email, application number, tool name, and request status. This structure helps organize tool request data in a systematic way.

Several custom objects are created to manage important information such as student, tool, and tool request. Each object contains fields that capture relevant data related to the system. The student object stores registration details, the tool object maintains information about available laboratory tools, and the tool request object records the requests submitted by students.

The database also defines relationships between objects to connect students with their tool requests and link tool requests with available tool records. Salesforce provides role-based access control to ensure that only authorized users such as lab incharge and head of department can review and update request records. This database design ensures that tool request information is stored securely and can be easily accessed and managed.

Overall, the database design plays an important role in the effective functioning of the lab tool request management system. By using salesforce objects, fields, and relationships, the system organizes tool request data in a clear and structured manner, ensuring efficient management of laboratory tools and request approvals. This also helps maintain accurate records and supports smooth system operation. It also improves data accessibility and management within the system.

3.8 TABLE DESIGN

Table Design refers to the process of organizing data into structured tables in a database. Each table consists of rows and columns, where rows represent records and columns represent fields. Table design defines the field names, data types, and relationships between different tables. A well-designed table helps store and retrieve data efficiently and ensures data accuracy, consistency, and easy management of information.

In the lab tool request management system, table design is implemented using salesforce objects, which function similar to database tables. These objects store important information related to the system such as student details, tool information, and tool request records. Each object contains fields that store specific data such as student name, email, application number, tool name, quantity requested, request date, and approval status.

The tables are also connected through relationships, which help link student records with their tool requests and tool details. This structure allows the system to retrieve and manage data efficiently. Overall, proper table design helps maintain organized data storage and supports the smooth functioning of the Lab Tool Request Management System.

3.8.1 STUDENT TABLE

Field Name	Data Type	Description
Student Name	Text	Name of the student
Application Number	Text	Unique application number
Email	Email	Student email address
Course	Picklist	Course or department of the student

Table 3.1 Student Table

In **Table 3.1**, the Student table stores information about students in the Lab Tool Request Management System. It contains details such as student name, application number, email, and course. This table helps the system maintain student registration records and identify each student who submits a tool request. It also allows the system to link the student with their tool request records for tracking request status and managing laboratory tool usage.

3.8.2 TOOL TABLE

Field Name	Data Type	Description
Tool Code	Text	Unique code for the laboratory tool
Tool Name	Text	Name of the laboratory tool
Quantity Available	Number	Number of tools available in the laboratory

Table 3.2 Tool Table

In Table 3.2, the tool table stores details of laboratory tools such as tool code, tool name, quantity, lab name, and description in the lab tool request management system. This table allows the Lab Incharge to check tool availability before approving requests and helps the system update tool quantity when tools are issued, ensuring proper management of laboratory tools.

3.8.3 TOOL REQUEST TABLE

Field Name	Data Type	Description
Request Number	Auto Number	Unique number generated or each tool request
Application Number	Lookup (Student)	Selected tool requested by the student
Tool Name	Lookup (Tool)	Selected tool requested by the student
Quantity Requested	Number	Date when the tool request was submitted
Status	Picklist	Status of the request

Table 3.3 Tool Request

In Table 3.3, the Tool Request Table stores details of tool requests submitted by students, such as application number, tool name, quantity, request date, and request status. It helps the lab incharge and head of department review and manage requests and allows the system to track tool request records and approval status.

SUMMARY

This chapter explains the design of the lab tool request management system using DFD, use case, and ER diagrams. It also describes the student, tool, and tool request tables used to manage system data.

CHAPTER 4

IMPLEMENTATION

4.1 CODE DESCRIPTION

Code description is important for explaining the functionality and structure of the source code used in the tool request management system. The project is implemented using the salesforce platform, which includes lightning web components (LWC) for the frontend interface, Apex for backend logic, and Salesforce objects and automation tools for managing data and workflows. Each module of the system is developed carefully to ensure clarity, modularity, and easy maintenance.

The code structure follows the standard salesforce DX project format, where components and classes are organized into specific folders. Lightning Web components are used to create the user interface for features such as student registration, tool request submission, and my tool request tracking. Each component contains separate files for HTML markup, JavaScript logic, and configuration metadata. Apex classes handle server-side operations such as retrieving tool data, storing student registrations, and processing tool requests.

Comments are added in the code to explain important logic, database interactions, and system processes. They help developers understand how different parts of the system work together and make the code easier to read, maintain, and modify. Proper commenting also helps during debugging and future system updates.

It also helps new developers understand the system quickly. Comments improve the clarity and structure of the program. Overall, they support better maintenance and development of the system. They explain important parts of the code clearly. This makes debugging and updating the system easier. It also improves the overall quality of the software.

4.1.1 STUDENT REGISTRATION LWC

The student registration lightning web component is responsible for collecting and storing student registration details in the system. This component provides an online registration form where students enter their basic information such as name, email, and course. The component communicates with an Apex controller met head of department to store the entered data in the salesforce database and generate a unique application number for the student.

The component uses JavaScript event handling to capture the values entered in the form fields. These values are stored in variables and passed to the Apex met Head of Department when the student clicks the submit button. The apex met head of department processes the data and creates a new student record in the system. After successful registration, the generated application number is returned and stored in the browser using localstorage or sessionstorage so it can be used for submitting tool requests.

The HTML template displays input fields and a submit button for the registration form. When the registration is successful, the system shows a confirmation message and allows the student to proceed to the tool request page. This component ensures accurate data collection and acts as the first step of interaction between the student and the tool request management system.

```
handleSubmit() {
  registerStudent({
    name: this.name,
    email: this.email,
    course: this.course
  })
  .then(result => {
    this.applicationNumber = result;
    sessionStorage.setItem(
```

```
        'applicationNumber', result);
    })
    .catch(error => {
        this.error = error;
    });
}
```

4.1.2 TOOL REQUEST LWC

The tool request lightning web component is responsible for collecting and submitting tool request details from students. This component uses an imperative Apex call to send the request data to the server when the student clicks the submit button. Unlike the wire adapter, this met head of department executes only when the action is triggered by the user.

The component collects values such as application number, selected tool, and quantity from the input fields using javascript event handlers. These values are stored in variables and passed to the apex met head of department create request when the student submits the form. A loading indicator can be displayed while the request is being processed to improve the user experience.

Once the request is successfully submitted, the system stores the request record in salesforce and displays a confirmation message to the student. If any error occurs during the submission process, the system shows an appropriate error message. This component ensures that tool requests are submitted correctly and forwarded for the lab incharge and head of department approval process.

```
handleSubmit() {
    this.isLoading = true;
    createRequest({
        applicationNumber: this.applicationNumber,
```

```
        toolId: this.selectedToolId,
        quantity: parseInt(this.quantity)
    })
    .then(result => {
        this.requestId = result;
        this.showSuccess = true;
    })
    .catch(error => {
        this.error = error.body.message;
    })
    .finally(() => {
        this.isLoading = false;
    });
}
```

4.1.3 TOOLREQUEST CONTROLLER APEX

The create request met head of department in the tool request controller apex class handles the creation of tool request records submitted by students. This met head of department receives input values such as the application number, tool id, and quantity from the lightning web component and stores the request details in the salesforce database. It acts as the server-side logic that processes tool request submissions.

The met head of department follows salesforce best practices by creating a new record of the Tool_Request__c object and inserting it into the database. It links the request with the corresponding student record using the Application_Number__c field. After the record is inserted successfully, the system stores the request details and makes them available for the lab incharge and head of department approval process. The met head of department is annotated with @AuraEnabled, allowing the lightning web component to call

the apex met head of department. This ensures smooth communication between the frontend interface and the salesforce backend, enabling the system to process tool requests efficiently.

```
@AuraEnabled
public static Id createRequest(String applicationNumber,
                               Id toolId, Integer quantity) {
    Tool_Request__c req = new Tool_Request__c(
        Application_Number__c = applicationNumber,
        Tool__c = toolId,
        Quantity__c = quantity,
        Request_Date__c = Date.today(),
        Status__c = 'Pending'
    );
    insert req;
    return req.Id;
}
```

4.1.4 RECORD-TRIGGERED FLOWS IMPLEMENTATION

Record-Triggered Flows are used in the tool request management system to automate important steps in the tool request and approval process without writing additional code. These flows automatically perform tasks such as sending email notifications, updating request status, and forwarding requests to the next level of approval. They help reduce manual work and improve system efficiency. The flows ensure that the approval process follows the correct sequence.

Flow 1 is triggered when a new tool request record is created by a student. The flow automatically sends an email notification to the lab incharge informing them that a new tool request has been submitted and is waiting for review.

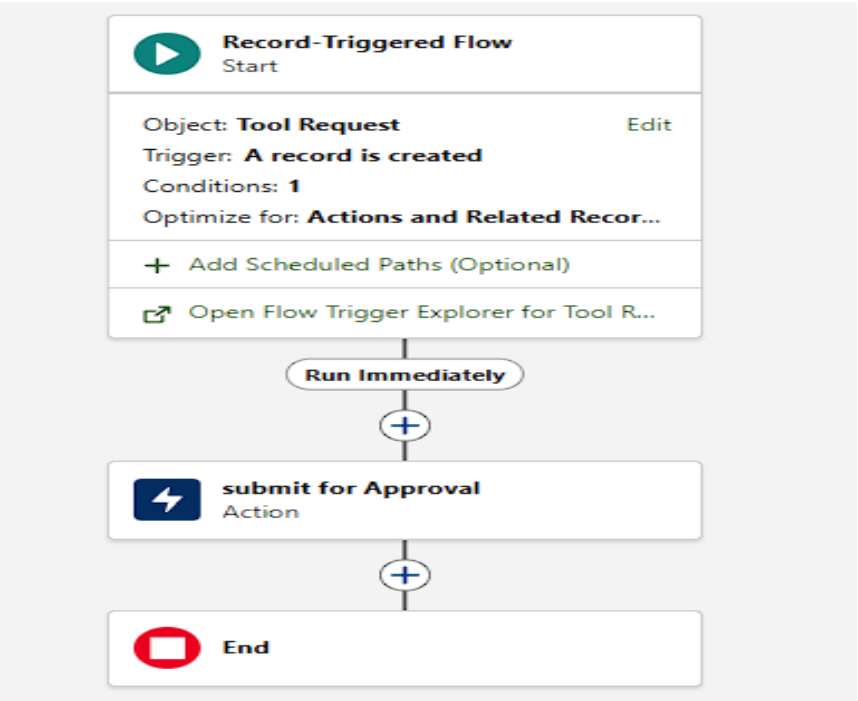


Figure 4.1 Triggered flow Diagram

Figure 4.1, flow 2 is triggered when the Lab Incharge approves the tool request. The flow updates the request status and automatically sends an email notification to the head of department for final approval.

Flow 3 is triggered when the head of department approves or rejects the request. The flow updates the final request status in the system and ensures that the request record reflects the latest decision.

All flows are designed using standard salesforce flow builder elements such as get records, update records, decision, and send email alerts. Since these flows do not require custom code, they can be easily maintained and modified by salesforce administrators when needed.

4.2 STANDARDIZATION OF THE CODING

Coding standards are followed throughout the development of the Tool Request Management System to maintain consistency, readability, and scalability of the application. Since the project is developed using the Salesforce platform, standard development practices are applied while creating custom objects, fields, Lightning Web Components, Apex classes,

and automation flows. Meaningful naming conventions are used for objects such as Student, tool, and tool request so that each component of the system can be easily identified and managed. Proper organization of components, fields, and automation processes helps keep the system structured and easy to understand.

Following standardized development practices also ensures that the system remains reliable and easy to maintain in the future. Lightning Web Components are organized with separate HTML, JavaScript, and metadata files, while Apex classes are written with clear met head of department s and secure coding practices. Salesforce Flows are designed in a simple and modular way to automate tasks such as sending email notifications and managing the approval process. Validation rules and appropriate field types are used to maintain data accuracy and prevent incorrect entries. Proper comments are also included in the code to explain important logic and processes. This standardized approach improves system reliability and ensures the smooth functioning of the tool request management system.

4.3 ERROR HANDLING

Error handling is an important part of the tool request management system to ensure smooth system operation and prevent incorrect data entry. The system includes validation checks to make sure users provide complete and correct information. Required fields such as student name, email, application number, tool selection, and quantity must be filled before submitting the form. salesforce validation rules help prevent invalid or incomplete data from being stored in the database. Proper field data types ensure that values like email and numbers follow the correct format. If users enter incorrect information, the system displays clear error messages. These messages guide users to correct the data before submission. Automation flows also help manage the request approval process and update request status. The system also prevents duplicate or invalid requests from being submitted. Overall, proper error handling improves data accuracy and system reliability.

SUMMARY

This chapter explains the implementation of the tool request management system using the salesforce platform. It describes the coding approach and the use of validation rules and flows to ensure accurate data and smooth system operation.

CHAPTER 5
TESTING AND RESULTS

5.1 TESTING

Testing is an important phase in the development of the tool request management system to ensure that the system functions correctly and meets the required objectives. It involves verifying that all modules such as student registration, tool request submission, my tool request, lab incharge approval, head of department approval, and reports are working properly without errors. Testing helps identify problems in the system and ensures that each component performs its intended function effectively.

In this project, testing is performed to verify whether students can register, submit tool requests, and track their request status through the system. It also ensures that the information entered by users is stored correctly in the salesforce database and that lab incharge and head of department can review and approve the requests properly. Through proper testing, the reliability, accuracy, and performance of the system are ensured before it is fully implemented. Testing also helps identify and fix errors in the system during the development phase. It ensures that each module of the system functions correctly and interacts properly with other modules. This process improves the overall quality and stability of the application.

Testing also helps ensure that system workflows and approvals operate smoothly. It verifies that email notifications and status updates work correctly. This process ensures the system performs efficiently under real user conditions. It also helps identify system errors. This improves the reliability of the application. Testing also confirms that all modules are properly integrated. It ensures that the system works correctly for all users.

The testing techniques applied in this project include:

- Unit Testing
- Integration Testing
- User Acceptance Testing

5.1.1 UNIT TESTING

Unit testing is performed to test individual modules of the tool request management system separately. Each module such as student registration, tool request submission, my tool request, lab incharge approval, head of department approval, and reports is tested individually to ensure that it performs its intended function correctly. This type of testing helps identify errors within specific components before integrating them with other modules and ensures that each module works independently. It also helps verify that each function works as expected. This improves the reliability of the overall system.

CASE 1

Module: Student Registration Module
Test Type: Student Registration Verification

Input: Student Name, Email, Course
Output: Student registration record created with Application Number

TEST
Input: Student enters valid registration details and submits the form
Output: Student record created and Application Number generated

Analysis: The registration details are stored correctly and a unique Application Number is generated for the student.
Result: Pass

CASE 2

Module: Tool Request Submission Module
Test Type: Tool Request Creation

Input: Application Number, Tool Name, Quantity
Output: Tool Request Record Created

TEST

Input: Student submits a valid tool request

Output: Tool request record created in the Tool Request object

Analysis: The request data is saved correctly and linked with the student record.

Result: Pass

CASE 3

Module: My Tool Request Module

Test Type: Tool Request Status Display

Input: Application Number

Output: List of submitted tool requests displayed with status

TEST

Input: Student enters Application Number to view requests

Output: System displays tool request details with status (Pending / Approved / Rejected)

Analysis: The system correctly retrieves and displays tool request records from the database.

Result: Pass

CASE 4

Module: Lab Incharge Approval Module

Test Type: Tool Request Verification

Input: Tool request details submitted by student

Output: Request approved or rejected by Lab Incharge

TEST

Input: Lab Incharge reviews the request and selects approve or reject

Output: Request status updated and forwarded to Head of Department if approved

Analysis: The approval process works correctly and updates the request status.

Result: Pass

CASE 5

Module: Head of Department Approval Module
Test Type: Final Approval Verification

Input: Tool request approved by Lab Incharge
Output: Final approval or rejection by HEAD OF DEPARTMENT

TEST
Input: Head of Department reviews the request and selects approve or reject
Output: Final request status updated in the system

Analysis: The system correctly updates the final decision of the request.
Result: Pass

CASE 6

Module: Reports and Monitoring Module
Test Type: Report Generation Verification

Input: Tool request and student data stored in the system

TEST
Input: Administrator views tool request reports
Output: Reports display request details and approval status

Analysis: The reports correctly retrieve data from the system and display accurate information.
Result: Pass

5.1.2 INTEGRATION TESTING

Integration testing is used to verify that different modules of the tool request management system work together correctly. in this project, it ensures that when a student registers and submits a tool request, the system correctly triggers the approval workflow and sends email notifications to the lab incharge and head of department. This testing confirms that the interaction between modules functions smoothly and that data flows correctly

between modules such as student registration, tool request submission, my tool request, lab incharge approval, head of department approval, and reports. Integration testing also verifies that tool request details are correctly stored in the salesforce database and updated across different stages of the approval process.

Integration testing helps identify issues that may occur when different modules interact with each other. It ensures that automation flows, approval processes, and data relationships function properly throughout the tool request workflow. This testing improves the reliability of the system and ensures proper coordination between all components of the tool request management system.

CASE 1

Modules: Student Registration Module → Tool Request Submission Module
Test Type: Module Interaction Verification

Input: Student registers and submits a tool request

TEST
Input: Valid registration details and tool request submitted
Output: Tool request record created and linked with the registered student

Analysis: The tool request submitted by the student is successfully stored and connected with the student record.
Result: Pass

CASE 2

Modules: Tool Request Submission Module → Lab Incharge Approval Module
Test Type: Automation Trigger Verification

Input: Student submits a valid tool request

TEST
Input: Tool request record created in the system
Output: Email notification sent automatically to Lab Incharge

Analysis: The system correctly sends a notification to the Lab Incharge after a tool request is submitted.
Result: Pass

CASE 3

Modules: Lab Incharge Approval Module → Head of Department Approval Module
Test Type: Workflow Progress Verification

Input: Lab Incharge approves the tool request

TEST
Input: Tool request status updated to approved by Lab Incharge
Output: Request Forwarded to Head of Department for final approval

Analysis: The system correctly forwards the approved request to the Head of Department.
Result: Pass

CASE 4

Modules: Head of Department Approval Module → My Tool Request Module
Test Type: Status Update Verification

Input: Head of Department Approves or rejects the tool request

TEST
Input: Final decision made by Head of Department
Output: Request status updated and visible in My Tool Request page

Analysis: The system correctly updates and displays the final request status.
Result: Pass

CASE 5

Modules: Tool Request Records → Reports and Monitoring Module
Test Type: Data Retrieval Verification

Input: Tool request and approval data stored in the system

TEST
Input: Administrator views tool request reports
Output: Reports display tool request details and approval status

Analysis: The reports correctly retrieve and display stored data from the system.
Result: Pass

5.1.3 VALIDATION TESTING

Validation testing ensures that the tool request management system follows the defined business rules and system requirements. It checks whether the system accepts valid inputs such as correct registration details, tool selection, quantity, and email format. This testing also confirms that the approval workflow functions correctly, where tool requests are first reviewed by the Lab Incharge and then forwarded to the head of department for final approval.

CASE 1

Module: Student Registration Module
Test Type: Required Field Validation

Input: Student submits registration form with missing fields

TEST
Input: Student leaves required fields empty and submits the form
Output: System displays an error message requesting the missing information

Analysis: The system correctly prevents incomplete registration submission.
Result: Pass

CASE 2

Module: Tool Request Submission Module
Test Type: Application Number Validation

Input: Invalid or missing Application Number

TEST
Input: Student attempts to submit request without a valid Application Number
Output: System displays an error and prevents request submission

Analysis: The validation rule correctly ensures that only registered students can submit **tool** requests.
Result: Pass

CASE 3

Module: Student Registration Module
Test Type: Email Format Validation

Input: Invalid email format entered

TEST
Input: Email value such as "invalid email"
Output: System displays an email format error

Analysis: The system correctly validates email format and prevents invalid data from being stored.
Result: Pass

CASE 4

Module: Tool Request Submission Module
Test Type: Quantity Validation

Input: Invalid quantity value

TEST
Input: Student enters negative or zero quantity
Output: System displays validation error and prevents submission

Analysis: The system correctly validates quantity values before storing the request.
Result: Pass

CASE 5

Module: Lab Incharge Approval Module
Test Type: Approval Workflow Verification

Input: Tool request submitted by student

TEST
Input: Lab Incharge reviews and approves the request
Output: Request forwarded to Head of Department for final approval

Analysis: The system correctly follows the defined approval workflow**.**
Result: Pass

CASE 6

Module: Reports and Monitoring Module
Test Type: Data Display Verification

Input: Tool request data stored in the system

TEST
Input: Administrator opens tool request reports
Output: Reports display request details and approval status

Analysis: Reports correctly retrieve and display stored data from the system**.**
Result: Pass

SUMMARY

This chapter discussed the testing and results of the tool request management system to ensure that the system works correctly and meets the specified requirements. Different testing met head of departments such as unit testing, integration testing, and validation testing were performed to verify the functionality of the system modules.

CHAPTER 6
CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The tool request management system developed using the salesforce platform provides an efficient and automated solution for managing laboratory tool requests in educational institutions. The system simplifies the tool request workflow using salesforce technologies such as lightning web components, apex, experience cloud, flows, reports, and dashboards.

It replaces the traditional manual met head of department of requesting laboratory tools with a centralized digital system, improving data accuracy and reducing administrative workload. The system allows students to register, submit tool requests, and track their request status through an online portal.

The system also enables lab incharge and head of department to review and approve tool requests through a structured approval process. Automated features such as email notifications, approval workflows, and status updates ensure that the tool request process runs smoothly. All request data is stored in Salesforce objects, making it easier to track and manage tool usage.

In addition, this project provided valuable experience in developing a cloud-based application using the Salesforce platform. It helped in understanding important concepts such as data modeling, Lightning Web Component development, Apex programming, automation using flows, and report generation. The system can be further enhanced in the future by adding features such as tool inventory tracking and mobile accessibility.

6.2 FUTURE ENHANCEMENT

The tool request management system can be further improved in the future by adding additional features that enhance its functionality and user experience. One possible enhancement is the implementation of a tool inventory management feature that allows administrators to track the total number of tools available, tools issued, and tools returned. This will help maintain accurate records of laboratory tools and ensure better resource management. Another improvement could be the addition of a tool return tracking system, where students can update the status after returning the borrowed tools. This will help maintain proper tracking of laboratory equipment.

Additional enhancements may include the development of a mobile-friendly interface or mobile application so that students and laboratory staff can access the system easily through smartphones. The system can also include automated notifications or reminders to inform students about tool request approvals, rejections, or return deadlines. Multi-language support can also be introduced to make the portal more accessible to students who prefer regional languages. Improved reporting and dashboard features can help administrators analyze tool usage and request trends more effectively. The system can also include search and filter options to easily find specific tool requests. A notification panel can be added to show important updates to users. These improvements will enhance the overall usability of the system.

Another possible enhancement is the integration of the system with other institutional systems, such as student management systems or laboratory inventory systems. This integration would allow automatic verification of student information and better management of laboratory resources. Additional security features such as advanced authentication and secure data storage can also be implemented to improve data protection. The system can also include role-based access control to restrict access based on user roles. regular system updates and monitoring tools can help maintain system performance. These improvements will make the system more scalable, secure, and efficient for use in educational institutions.

APPENDICES

A. SAMPLE CODING

TOOL REQUEST CONTROLLER

```
public without sharing class ToolRequestController {

@AuraEnabled

public static Map<String, String> registerStudent(String name, String email, String course)
{

 try {

 Student__c s = new Student__c(

 Name      = name,

 Email__c  = email,

 Course__c = course );

insert s;

Student__c stu = [

 SELECT Id, Application_Number__c

 FROM   Student__c

WHERE  Id = :s.Id

LIMIT  1  ];

 return new Map<String, String>{

  'studentId'        => (String) stu.Id,
```

```
'applicationNumber' => stu.Application_Number__c   };
 } catch (Exception e) {
 throw new AuraHandledException('Registration failed: ' + e.getMessage());
 }   }
 if (tool.Quantity_Available__c == null || qty > tool.Quantity_Available__c) {
 throw new AuraHandledException('Quantity exceeds available stock.');  }
 Tool_Request__c req = new Tool_Request__c(
  Student__c            = stu.Id,
  Student_Email__c      = stu.Email__c,
  Tool__c               = toolId,
  Quantity_Requested__c = qty,
  Requested_Date__c     = Date.today(),
   LIMIT  1     ];
     if (stuList.isEmpty()) {
 return new List<Tool_Request__c>()  }
 return [
 SELECT Id,
 Name,
 Quantity_Requested__c,
  Requested_Date__c,
```

```
Status__c,
Tool__r.Name
FROM   Tool_Request__c
WHERE  Student__c = :stuList[0].Id
 ORDER BY CreatedDate DESC
 @AuraEnabled(cacheable=true)
 public static List<Student__c> searchSuggestions(String keyword) {
   try {
  if (String.isBlank(keyword)) {
   return new List<Student__c>();  }
    return [
     update req;
     } catch (Exception e) {
     throw new AuraHandledException(e.getMessage());
 } }
```

TOOL REQUEST SERVICE.JS

```
import { LightningElement, api, wire } from 'lwc';
import getStudentRequests from '@salesforce/apex/ToolRequestController.getRequests';
export default class ToolRequestService extends LightningElement {
 @api appNumber;s
```

```
@wire(getStudentRequests, { appNumber: '$appNumber' })
 wiredRequests({ data, error }) {
  if (data) {
 console.log('Requests data:', data);
  } else if (error) {
   console.error('Error fetching requests:', error);
       }  }
```

STUDENT REGISTRATION.JS

```
import { LightningElement, track } from 'lwc';
import { NavigationMixin } from 'lightning/navigation';
import Toast from 'lightning/toast';
import registerStudent   from '@salesforce/apex/ToolRequestController.registerStudent';
import searchSuggestions from
'@salesforce/apex/ToolRequestController.searchSuggestions';
import verifyUser        from '@salesforce/apex/ToolRequestController.verifyUser';
import KEC_LOGO from '@salesforce/resourceUrl/kec_logo';
export default class StudentRegistration extends NavigationMixin(LightningElement) {
 logo = KEC_LOGO;
  // Mode: 'home' | 'login' | 'register' | 'search'
  @track mode = 'home';
```

```
// Guest flag
@track isGuest = false;
 get isHome()     { return this.mode === 'home'; }
 get isLogin()    { return this.mode === 'login'; }
 get isRegister() { return this.mode === 'register'; }
 get isSearch()   { return this.mode === 'search'; }
  showToast(label, message, variant = 'success') {
  Toast.show({ label, message, variant, mode: 'dismissible' }, this);  }
  /* HOME */
  handleLoginClick()  { this.mode = 'login'; }
  handleGuestClick() {
   this.mode        = 'search';  }
  setTimeout(() => {
  this[NavigationMixin.Navigate]({
  type: 'comm__namedPage',
    attributes: { name: 'Tool_Request__c' },
    state: {
    applicationNumber: result.applicationNumber,
    studentId        : result.studentId   }       });
```

B.SCREENSHOTS

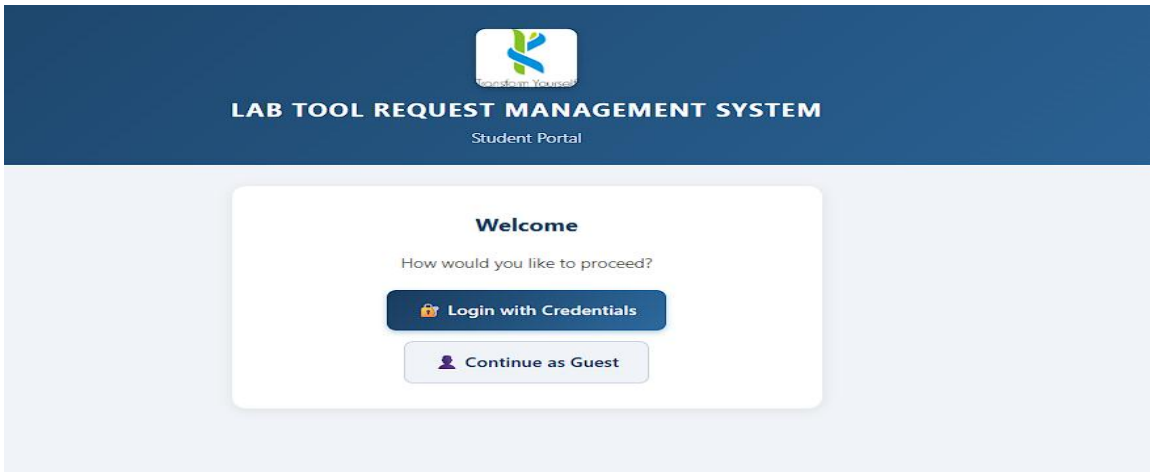


Figure B.1 Home Page

Figure B.1, this is the home page of the Lab Tool Request Management System where students can choose to log in with credentials or continue as a guest to access the student portal.

Figure B.2 Guest Home Page

Figure B.2, this is the guest home page where students can search their request using the application number or register as a new student to submit a tool request.

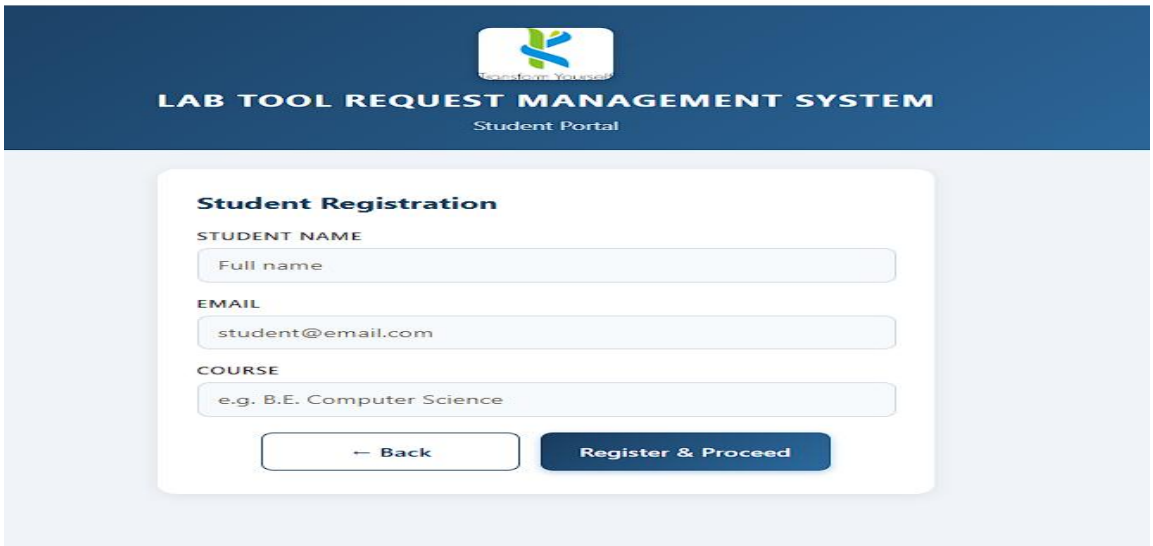


Figure B.3 Student Registration

Figure B.3, this is the student registration page where new students can enter their name, email, and course details to register and proceed with the tool request.

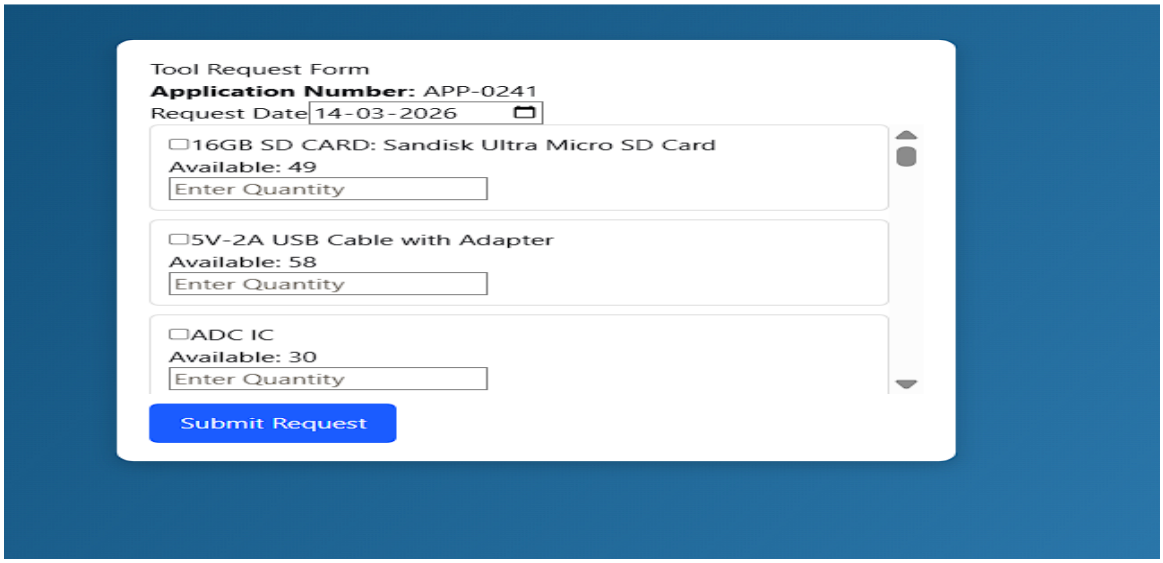


Figure B.4 Tool Request

Figure B.4 , this is the tool request page where students can view available lab tools, enter the required quantity, and submit their request using the application number.

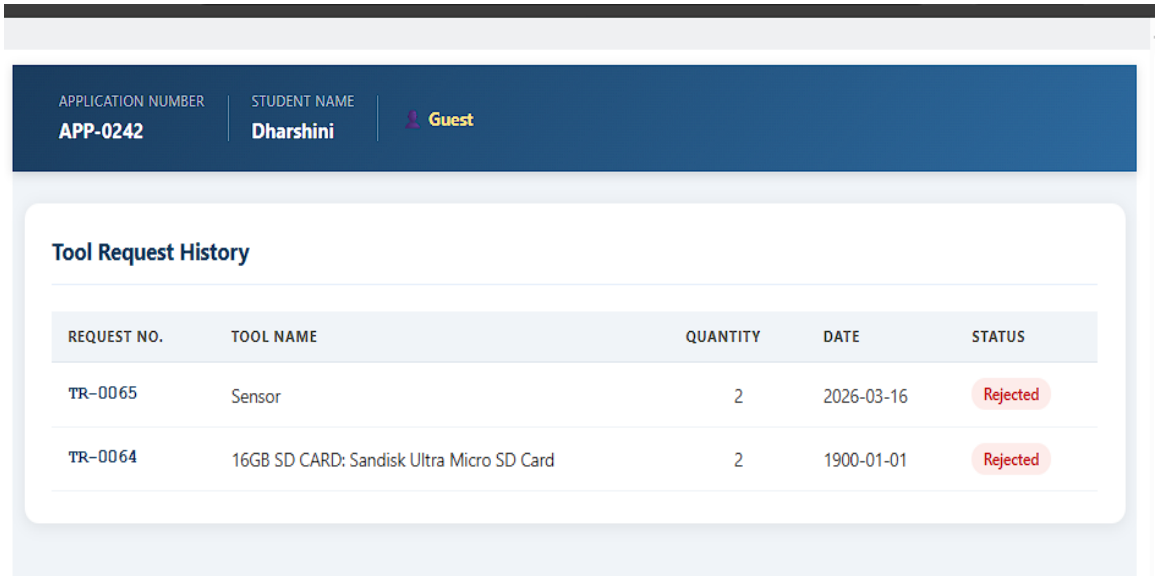


Figure B.5 Tool Request History

Figure B.5, this is the tool request history page where students can view their submitted requests along with tool name, quantity, date, and approval status.

Figure B.6 Tool Request Approval

Figure B.6 ,this is the tool request notification page where users receive the request details, status of approval, and a link to view the submitted tool request.

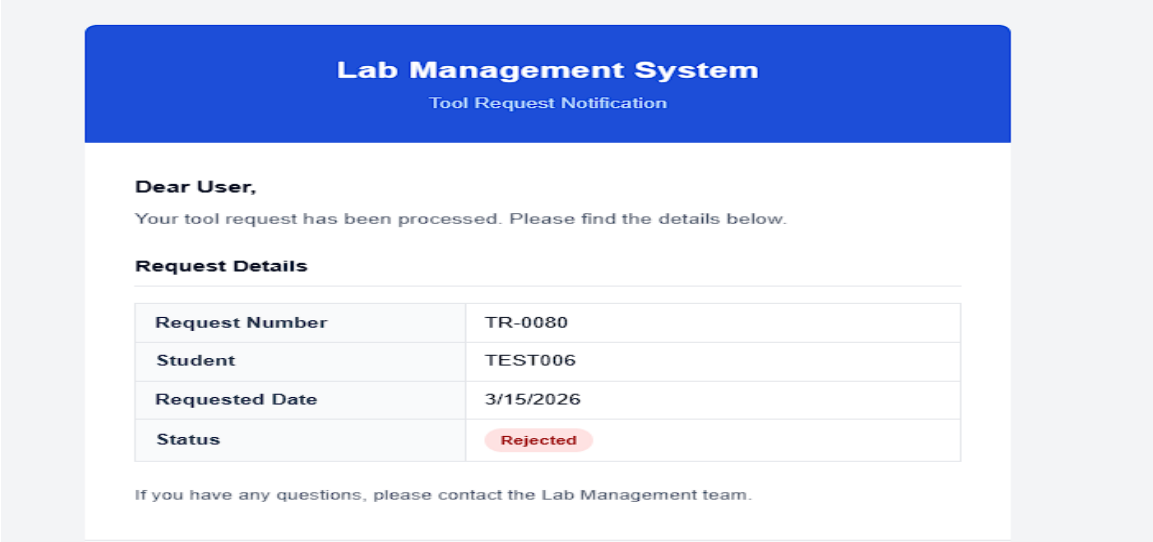


Figure B.7 Tool Request Rejected

Figure B.7 , this is the tool request notification page where users receive the details of their request and the system informs them that the tool request has been rejected.

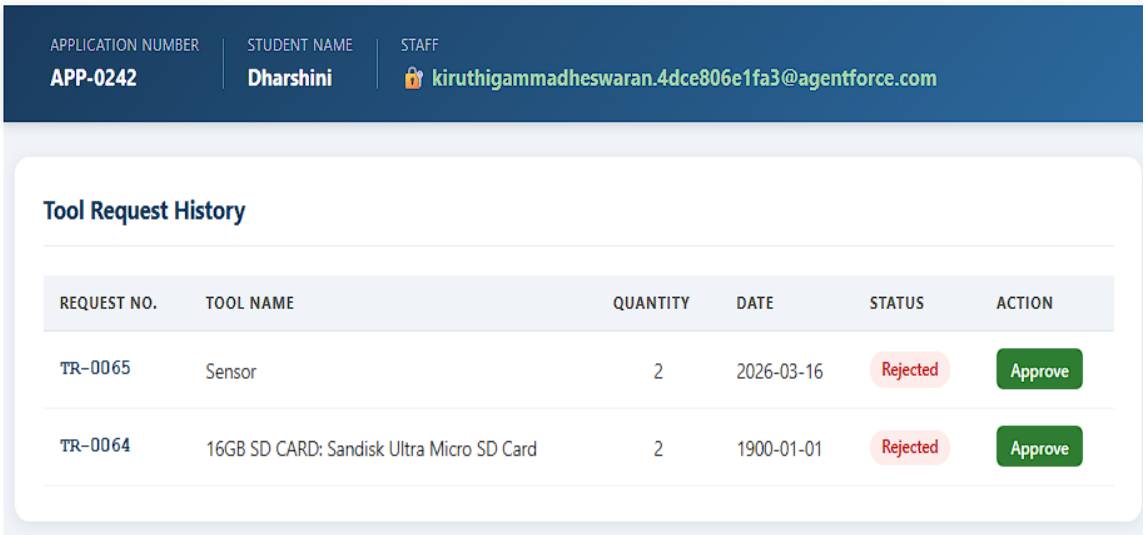


Figure B.8 Admin Tool Request History

Figure B.8, this is the staff dashboard page where the lab staff can view student tool requests, check the request details, and approve the requests.

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